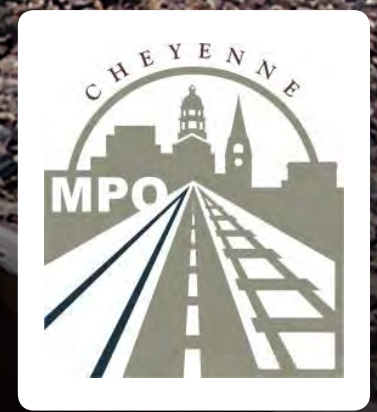


CROSSROAD



REED AVENUE RAIL CORRIDOR MASTER PLAN

APRIL 2018



DRAFT

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RAIL CORRIDOR
MASTER PLAN

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INTRODUCTION

Reed Avenue has been and will continue to be an important corridor within the West Edge District of Downtown Cheyenne.

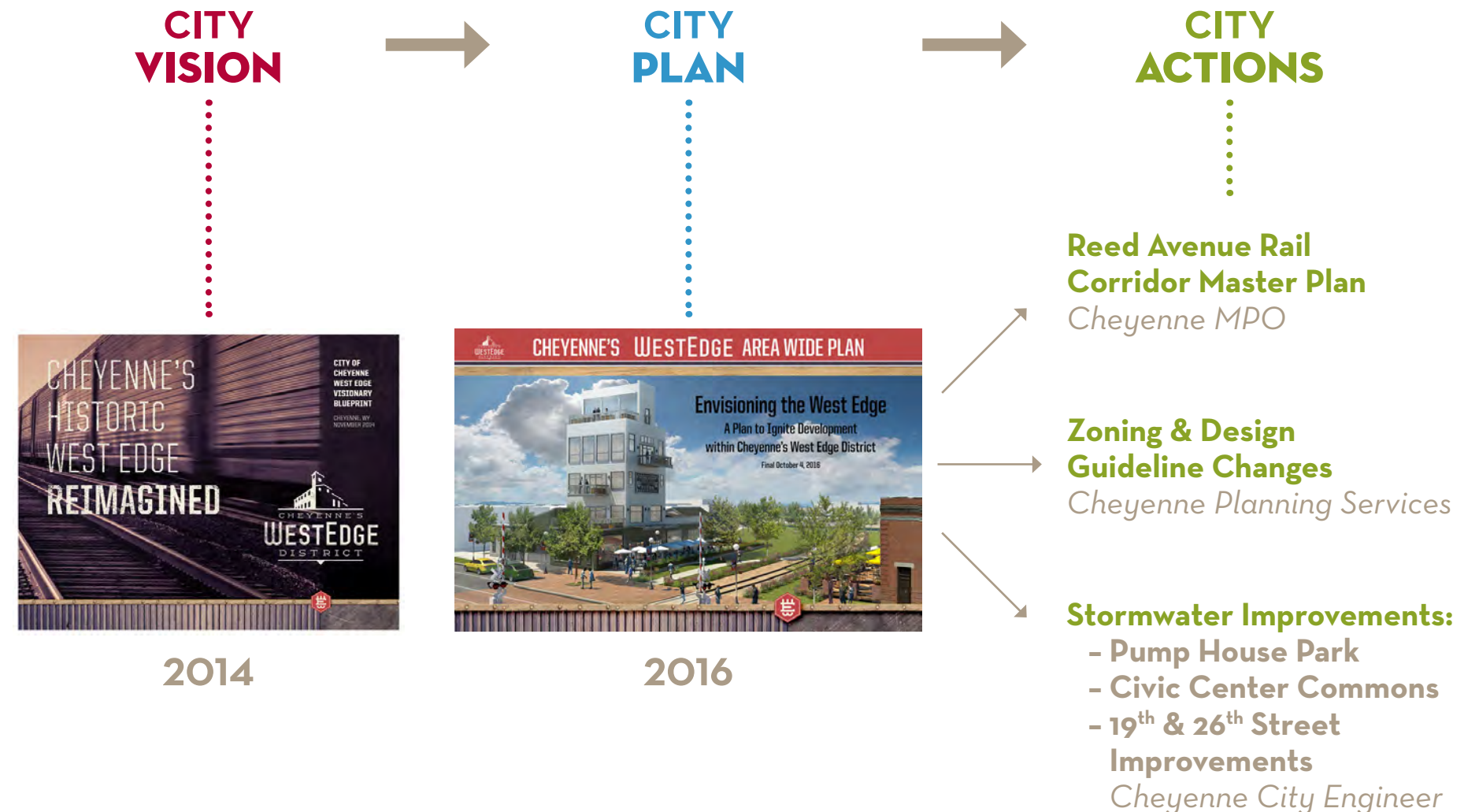
In the early twentieth century, the City of Cheyenne (City of Cheyenne, City, or Cheyenne) granted an easement to Burlington Northern Santa Fe (BNSF) to build and operate a rail corridor within Reed Avenue's 80-foot right-of-way. Rail service within Reed Avenue enabled many businesses to ship and receive goods via the railway. Properties adjoining the rail corridor became some of Cheyenne's first and more successful industrial land uses.

Unfortunately, over time, many of the industrial properties in the Reed Avenue Corridor declined as industrial activity and its dependency on rail service declined in preference for the trucking industry and locations closer to the State and Federal Highway Systems. Today, the West Edge District, once one of Cheyenne's leading industrial areas, is underperforming and in need of a new economic identity.

The City of Cheyenne has been studying Downtown Cheyenne, specifically its West Edge District's decline, for several years and has framed solutions for its revitalization. The City's *Historic West Edge Reimagined* (2014) established a 'Visionary Blueprint' for the District's revitalization, calling for it to shift from an underperforming, primarily industrial segment of Downtown into a more vibrant mixed employment, commercial, and residential district. The subsequent *West Edge Area Wide Plan* (2016) established an action plan for achieving the District's vision. It identified strategies for overcoming three key physical, psychological, and economic barriers to the district's revitalization: the railroad, seasonal flooding, and a lack of economic drivers. The City of Cheyenne is addressing each barrier identified with separate initiatives.

This Reed Avenue Rail Corridor Master Plan, managed by the Cheyenne Metropolitan Planning Organization (MPO), focuses on minimizing the revitalization barriers associated with the condition of the 80-foot right-of-way and is intent on positioning the Corridor for the return of economic prosperity. The City Engineer is addressing the barriers related to seasonal flooding through four stormwater improvement projects: Pump House Park; Civic Center Commons; and the 19th and 26th Streets drainage improvements. Finally, the City's Planning Services Department is proposing to rezone large portions of the district from light industrial to mixed-use business to allow both light industrial and more diverse land uses to emerge and diminish the regulatory barriers associated with the lack of economic drivers.

West Edge District Planning & Implementation Process



PURPOSE OF THE STUDY

The purpose of this Master Plan is to establish a community-supported design concept, cost estimates, and an action plan for Reed Avenue's 80-foot right-of-way so the street can contribute to, and not be perceived as burdening, the revitalization efforts of the West Edge District and Downtown Cheyenne.

The design objective for the effort is to develop a 30% design concept for Reed Avenue which:

1. Enables the Corridor to revitalize as a vibrant multi-use corridor;
2. Safely co-exists with rail operations; and,
3. Celebrates the West Edge's historic and industrial themes.

The study extends along the blocks adjoining Reed Avenue and the BNSF Corridor from Lincolnway (16th Street) and Pump House Park to 23rd Street. This plan supports the common vision regarding the redevelopment of Reed Avenue and the West Edge Avenue District as envisioned in the adopted City of Cheyenne's *West Edge Area Wide Plan*.

A 30% design is conceptual and not final. A 30% design establishes the horizontal, not vertical, controls to the streetscape effort along with estimated costs. The 30% design is the foundation upon which specific design decisions (60%, 90%, and 100%) for Reed Avenue are to be based. More specific design considerations, such as selecting tree types, specific lighting fixtures, or paving material, will emerge as the project advances through final design.

REPORT ORGANIZATION

This report is divided into four sections. The first section, **Process and Community Objectives**, outlines the planning and public engagement efforts of the study along with community objectives for Reed Avenue which emerged during this effort. The second section, **Context and Corridor Assessment**, summarizes the regulatory, economic, physical, and operational attributes of the Reed Avenue Rail Corridor and the challenges that need to be overcome for the project to be successful. The third section, **Conceptual Design**, illustrates the preliminary design concept and initial cost estimates for the 80-foot right-of-way's redesign. The final section, **Implementation Action Plan**, outlines a fiscal assessment of the recommended improvements along with specific utility and railroad coordination plans necessary to advance the Reed Avenue Rail Corridor Master Plan with BNSF and the Public Utility Commission. The appendix of the report includes important supporting market and fiscal analysis documents along with detailed engineering sheets which illustrate the conceptual design of the Corridor. ☰

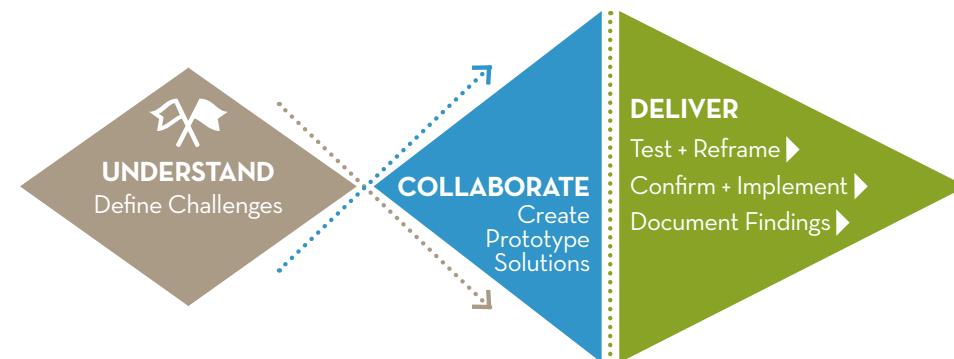


Reed Avenue Corridor Master Plan // Study Area

PROCESS & COMMUNITY OBJECTIVES

Meaningful public input is critical to the successful revitalization of the Reed Avenue Corridor.

Throughout the planning and public engagement effort, West Edge District community members worked alongside City staff to guide the project and ensure the Rail Corridor's redesign fit within the District's overall fabric. The project's three-phased planning and design effort was organized to ensure the Corridor stakeholders (property owners, including the railroads), business owners, and residents maintained an active voice throughout the project and could influence its outcome and take ownership in its subsequent implementation.



PHASE 1 – SHARED UNDERSTANDING

The first phase of the effort focused on developing a thorough understanding of the regulatory, physical, operational, and economic aspects of the Reed Avenue Rail Corridor. Stakeholder focus group meetings, a community walking audit, a project steering committee meeting, and a public workshop were used to ensure the Corridor's many vested stakeholders could share their insights about the Corridor's many challenges and how its opportunities could fit within the overall fabric of the West Edge District and the area's future prosperity.

A public kick-off meeting was held in the Corridor on April 14, 2017. Approximately 60 people attended this meeting. Participants were asked to identify: 1) what they liked most about the Corridor and what should be preserved; and 2) what they disliked in the Corridor and what they would like to see changed.

Participants most "liked" the Corridor's proximity to Downtown Cheyenne and its opportunity to revitalize. They also "liked" the historic architecture, the railroad, and the industrial character of Reed Avenue. Their "dislikes" centered around the physical conditions of the Corridor, its lack of amenities, and its abundance of underutilized properties.



Community Workshop #1, April 14th, 2017



Community "likes" and "dislikes" within the West Edge District and the Reed Avenue Rail Corridor.

PHASE 2 – COLLABORATIVE DESIGN

The second phase of the effort concentrated on **collaborating** with the community to develop a range of prototype solutions for the Reed Avenue right-of-way. A multiday design charrette, open to the public, was held June 20 – 22, 2017 in the Laramie County Library. Over 100 people participated over the three-day period.

The design charrette format was used for four reasons:

1. The charrette allowed those who influence the project to develop a vested interest in the design and take ownership in its implementation;
2. The charrette allowed the project team, the City, and the community to generate multi-disciplined ideas and discuss issues together, maximizing efficiency and productivity;
3. The charrette allowed a collective effort to organize all participants' input during focused meetings thereby eliminating the need for prolonged discussions that typically delay conventional planning projects; and,
4. The charrette allowed an evolution of ideas and design decisions to occur in a public forum, allowing members of the community see how their insights are translated into specific design strategies.

The Reed Avenue charrette started with an evening public meeting on June 20th. During the meeting, the community was asked to help identify specific goals and objectives for the Reed Avenue right-of-way's redesign. Participants eloquently communicated their expectations for the redesign of Reed Avenue to “**shift**” from the Corridor's many negative underutilized industrial conditions of the Corridor to more positive outcomes for the West Edge District capable of supporting a mix of land uses and a walkable urban environment.

The second day of the charrette was an open house for community members to come as they pleased. Conversations were one-on-one and informal. During the open house, the project steering committee held a meeting to discuss the project and the initial findings of the evening public workshop. The final day of the charrette concluded with a public presentation. Two functional design concepts for Reed Avenue were developed during the three-day design charrette. Both options introduced an improved pedestrian experience on each side of the existing BNSF rail corridor. The two concepts showcased the functional difference available to the City, not specific design details. Feedback received during the final charrette presentation were incorporated into the final recommended design for Reed Avenue in Phase 3.



Community Design Charrette, June 20 – 22, 2017

THE SHIFT

from

to

A place to drive though

Pioneers
No support for businesses or residences

Abandoned buildings

Barriers

Nobody lives here

Wide asphalt

Narrow sidewalks

Streets for cars

Hostile

Few open spaces

Kids can't enjoy

Nothing to do

Unsustainable



A place to live, work, and play

Place of investment

Many partners

New investment

Seams

Vibrant neighborhoods

Wider sidewalks

Multi-use corridors

Streets for people

Inviting

Great experiences

Kids can cross the street

Food and entertainment

Sustainable

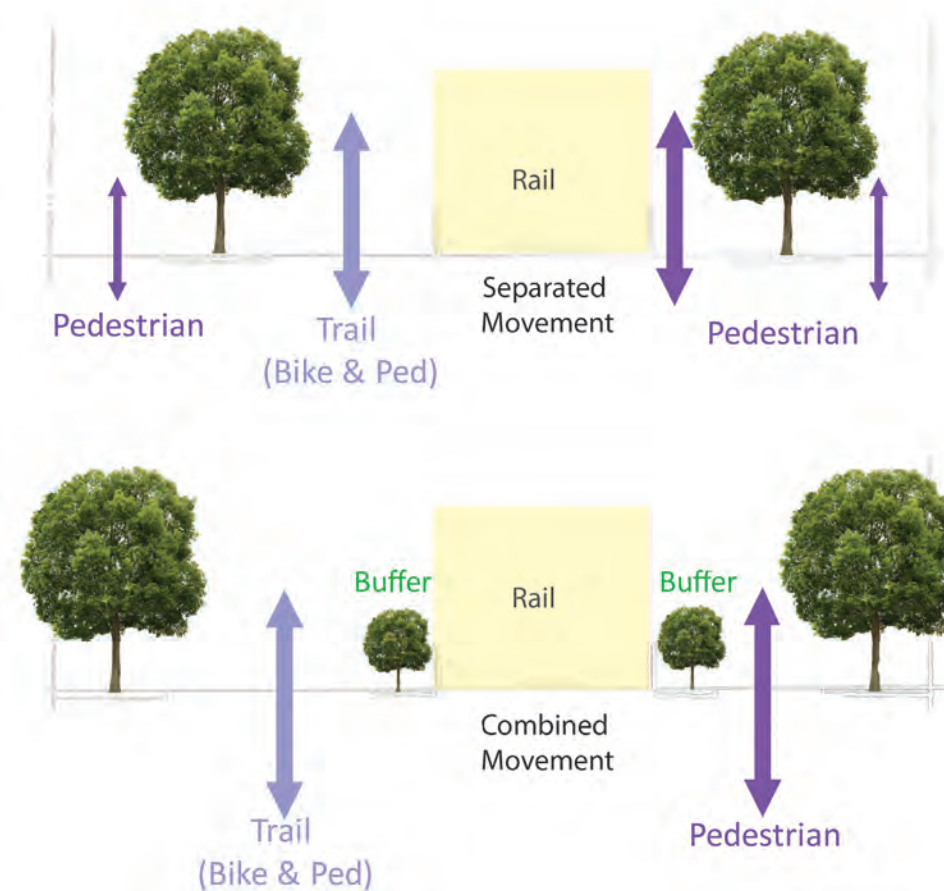
PHASE 3 – DELIVERY

The concept plans developed during the multi-day design charrette and vetted with the community were tested against defined community success measures. Refined cost estimates and initial project prioritization recommendations—based on economic opportunity assessments, costs, and financial resources—were developed and advanced for review, feedback, and confirmation from the project steering committee and the community.

A preliminary set of recommendations was developed and presented to the public for final confirmation on October 3, 2017. Approximately 80 people attended the final presentation, which was also broadcast through Facebook Live on the City's Facebook page. ☰



Project Steering Kick-off, April 14, 2017



Conceptual differences between alternatives developed during the design charrette.



Project Steering Committee meeting held during the design charrette; June 20, 2017

CONTEXT & CORRIDOR ASSESSMENT

This context assessment describes the conditions in the study area and key trends and challenges facing the Reed Avenue Rail Corridor. The Reed Avenue Rail Corridor, located in the western portion of Downtown Cheyenne, stretches along Reed Avenue and the BNSF rail corridor from Lincolnway north to 23rd Street. The study area includes the 80-foot right-of-way of Reed Avenue, the BNSF rail corridor, and the adjoining blocks between Bent Avenue and Snyder Avenue.

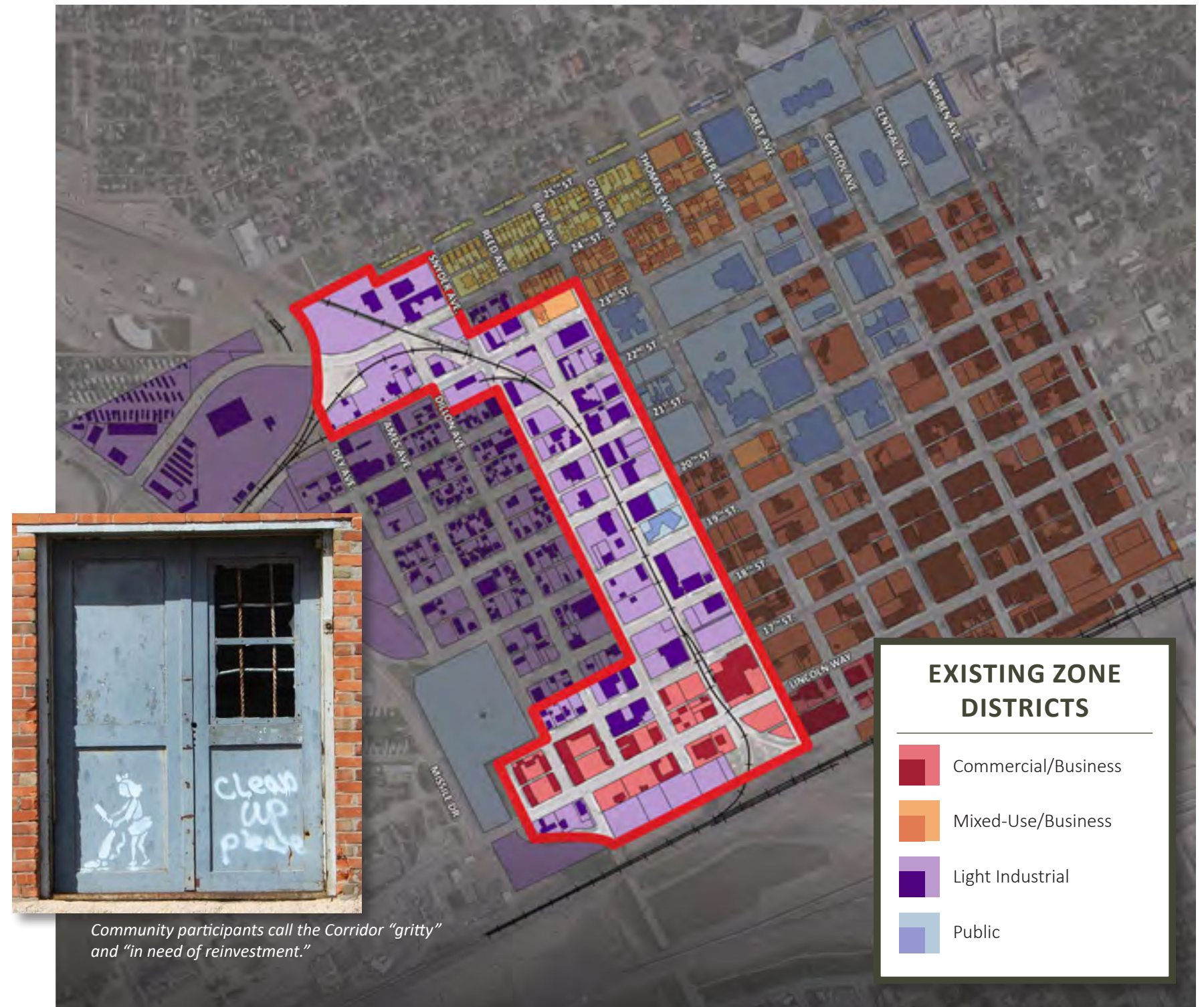
EXISTING LAND USES AND ZONING

The study area is approximately 90 acres in size. Nearly half (40 acres) of the study area's land area is dedicated to public streets and alley right-of-way. The developable land in the study area is approximately 50 acres in size. The existing land uses in the Corridor reflect the City's current zoning for the area: 67% light industrial (33.5 acres); 28% commercial business (14 acres); 3% public (1.5 acres); and 2% mixed-use business (1 acre).

The State of Wyoming, the City of Cheyenne, and the Laramie County School District #1 own approximately 16 acres, or 30% of the developable land in the study area. The majority of the publicly owned properties are zoned and operate as industrial uses. The exceptions are an institutionally-zoned fire station at 19th Street and Reed Avenue and school at 18th Street and Bent Avenue. Two properties near 23rd Street and Reed Avenue are zoned as mixed-use business.

Approximately two thirds of the study area is limited in its land use options. The light industrial zone district allows all employment-based and the majority of retail-based land uses. However, the light industrial zone district prohibits bars and restaurants along with all residential land uses. Additionally, the light industrial zone district does not require developments be designed to prioritize pedestrian activity.

For these reasons, the City is seeking to rezone the Corridor from a light industrial to a mixed-use business district. This change in zoning would allow the full range of employment, retail, and residential land uses as well as require new development to be designed to support a more walkable urban form—consistent with Downtown Cheyenne.



Current Zoning Districts

DISTRICT CHARACTER AND BUILDING STOCK

Reed Avenue was described by workshop participants as “gritty,” rich with rail history, and having a strong industrial character. The Reed Avenue Corridor was also described as “underutilized,” “not reaching its potential,” and “in need of reinvestment.” The Corridor’s building stock is eclectic, and in many ways matches the community’s observations and concerns about the Corridor.

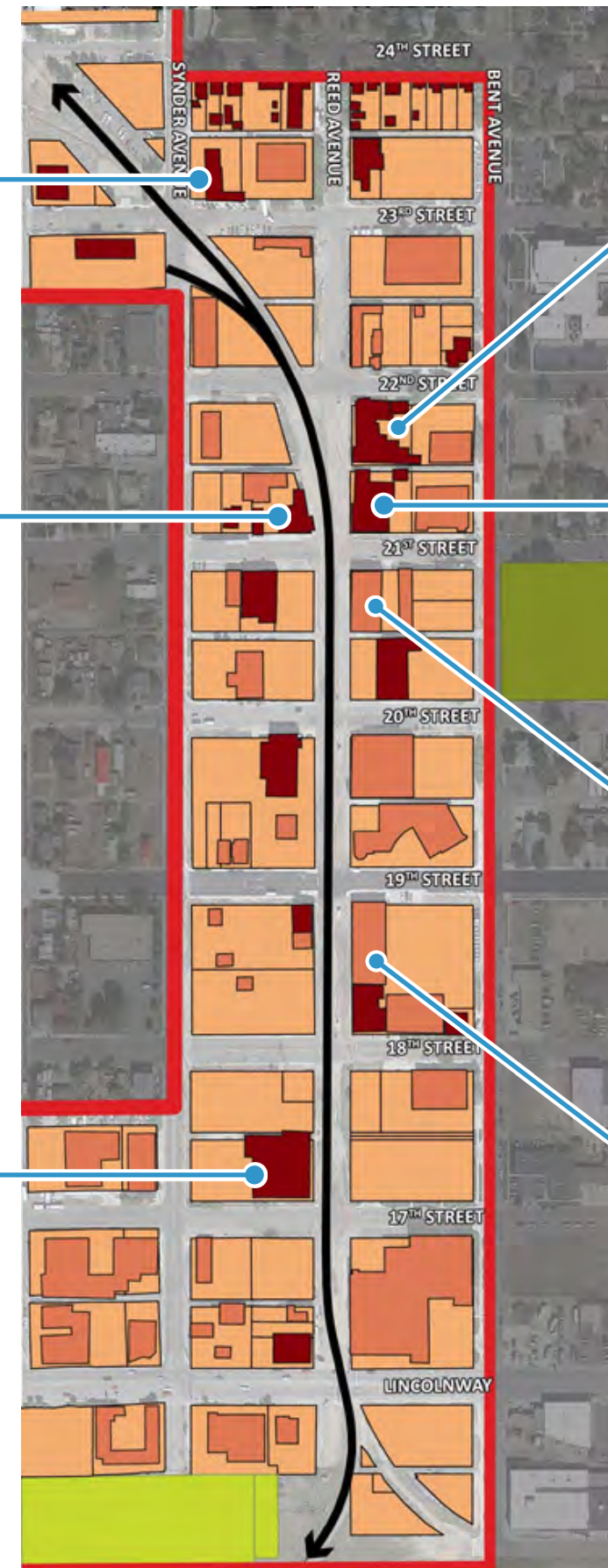
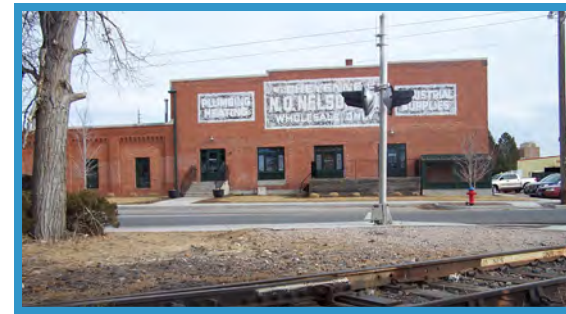
There are several character-rich buildings, primarily brick industrial structures, which reflect the industrial architecture of the past. Additionally, the grain elevator at 21st and Reed and the feed structure at 22nd and Reed are unique wood-framed structures that provide connections to the City’s agricultural history.

Each of these older buildings displays architectural materials and details that are unique and not seen in current building styles. Together, they invoke a rich sense of history that is distinctively Cheyenne. The materials from these structures—traditional steel, brick, and wood—were identified by the community as needing to be replicated in the redesign of the Reed Avenue Corridor right-of-way.

A few of these character-rich buildings have been renovated and adaptively reprogrammed for contemporary office and commercial uses. However, unfortunately, many of the other character-rich structures in the Corridor remain in a state of disrepair and in need of reinvestment.

A majority of the Corridor’s remaining building stock was built between 1970 and the early 2000s. Several of these newer buildings lack architectural details and have been designed for single-purpose industrial activities. Adaptive reuse of these structures will be more difficult and, in many instances, impractical. The buildings have limited flexibility as they are aluminum butler buildings or non-descript concrete masonry unit (CMU) structures designed for warehousing. Unlike many of the older brick buildings, these newer industrial buildings have economic value today as industrial properties and their redevelopment may not occur in the near future.

A goal for the Reed Avenue Rail Corridor Master Plan is to redesign the right-of-way so that the development industry is motivated to reinvest in the character-rich buildings and reprogram them into a contemporary mix of office, commercial, and residential uses alongside the successful industrial activities in the Corridor.



CORRIDOR BUILDING STOCK

- Character-Rich/Adaptable Structures
- Single-Purpose/Limited-Use Structure

THE STREET NETWORK, PROPERTY ACCESS, AND TRAFFIC CIRCULATION

Downtown Cheyenne and its West Edge District maintain a traditional interconnected street network which generates a 300-foot block structure. This traditional street grid allows tremendous flexibility for the ultimate redesign of the Reed Avenue Corridor.

Well-connected street networks distribute traffic and enable more transportation choices and more land use opportunities. Interconnected streets provide motorists with multiple routing options and relieve pressure to accommodate traffic movements along a single corridor.

The traditional street network of the West Edge District provides the Reed Avenue Corridor's redesign the opportunity to eliminate vehicular travel within the right-of-way with little or no impact to the City's traffic operations. Today, the unimproved Reed Avenue allows low-volume vehicle access, truck traffic, and pedestrian and bicycle movements to all coexist with the rail activity within an uncontrolled environment.

The primary vehicular traffic operation of Reed Avenue is to provide property access to the Corridor's adjoining properties, not vehicular mobility. Four blocks of Reed Avenue provide direct vehicle access to private properties. In reviewing access to these parcels, the properties on Reed Avenue between 19th and 21st Streets will need to be modified to ensure access can be provided via alternative connections to Reed Avenue. Simple modifications to each of these properties can cure access concerns so that Reed Avenue can be redesigned to prohibit vehicular traffic between 17th and 22nd Streets.

The parcels along Reed Avenue between Lincolnway and 17th Street and properties fronting Reed Avenue between 22nd and 23rd Streets on the north and south ends of the Corridor should not be converted to pedestrian- and bicycle-only traffic. Access to properties fronting these blocks will continue to require vehicular access from Reed Avenue in the Corridor's redesign.

Primary streets within the study area, arterials or collectors, which provide important vehicular circulation into and around Downtown Cheyenne include: Lincolnway; 19th Street; 20th Street; 24th Street; and Snyder Avenue. The redesign of Reed Avenue must allow these roadways to maintain a higher standard of vehicular mobility when compared to the other streets in the study area.

Lincolnway is a state highway and the primary access route in and out of Downtown Cheyenne, to and from the west. Previous concepts of Reed Avenue identified a desire to create a pedestrian and bicycle traffic corridor from 23rd Street across Lincolnway. In reviewing roadway geometrics and traffic operations, it was determined that the intersection of Reed Avenue and Lincolnway is not a good location for a pedestrian crossing. That crossing should be shifted west one block to Snyder Avenue.

DRAFT



The intersection of Reed Avenue and Lincolnway is located at the top of a vertical curve with poor sightlines from Lincolnway. A traffic signal at Snyder Avenue and Lincolnway is one block (300 feet) west of Reed Avenue. A 300-foot spacing between a traffic signal at Snyder Avenue and a pedestrian signal at Reed Avenue is below the City's and the State's recommended minimum spacing and is not recommended.

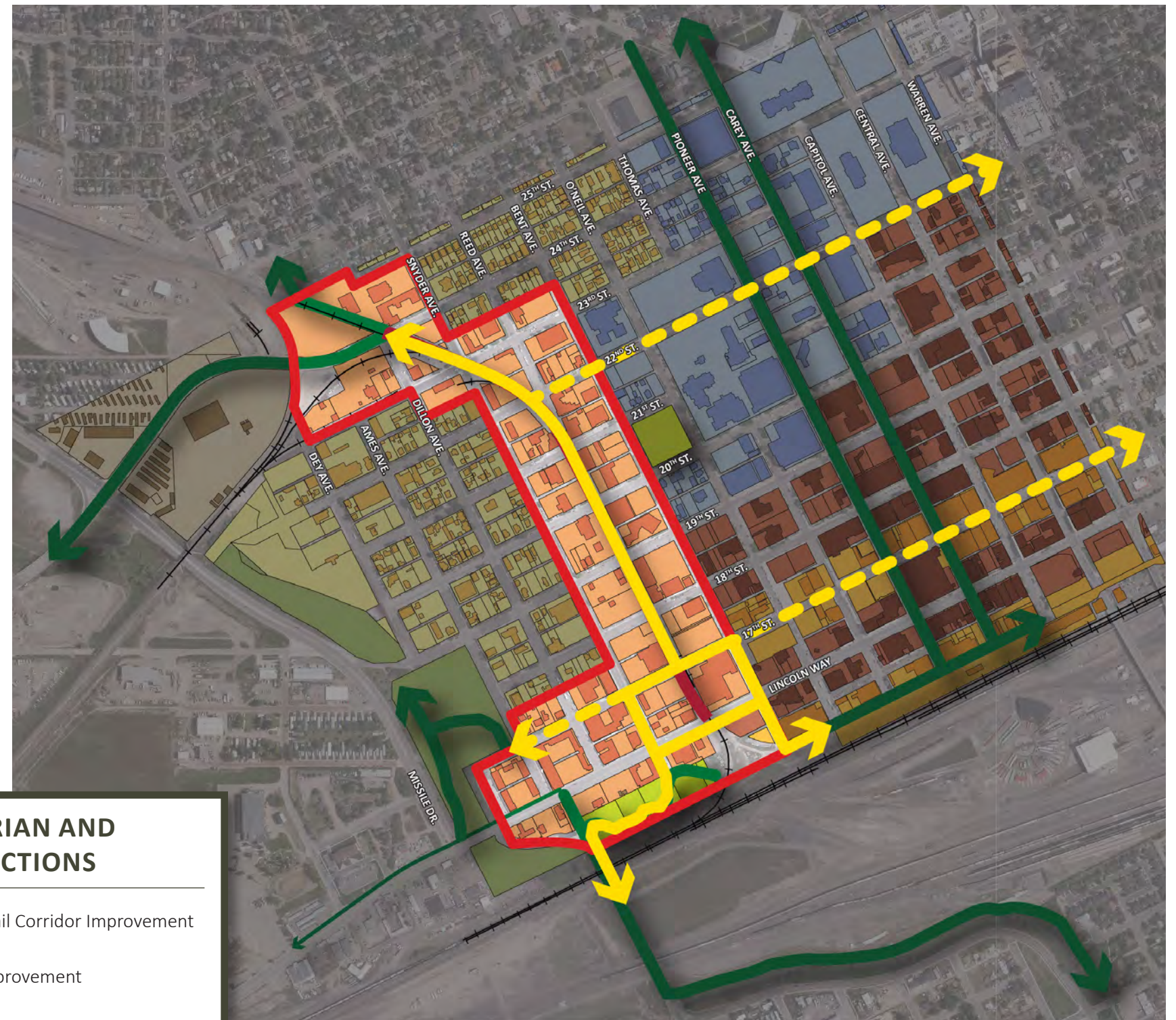
PEDESTRIAN AND BICYCLE FACILITIES

The City of Cheyenne has developed a high-quality city-wide bicycle network with an extensive greenway system encircling three quarters of the City. The Reed Avenue Rail Corridor presents a significant opportunity to complete the beltway around the City and significantly improve bicycle and pedestrian mobility in Downtown Cheyenne.

The gap in a city-wide greenway around Cheyenne is located Downtown. Recent City improvements to Pump House Park extend the City's greenway from the Ames Avenue underpass to the West Edge District and the Reed Avenue Corridor. However, there is a gap in bicycle facilities between Pump House Park and Carey Avenue and Pioneer Avenue Bike Lanes.

Introducing a greenway within the Reed Avenue Corridor with new bike facilities on 17th Street and 22nd Street presents an opportunity to complete the city-wide greenway and interconnect the key traffic generators and desired location within Downtown Cheyenne to the rest of the City on pedestrian friendly, multimodal corridors.

The connection of the city-wide greenway between Pump House Park and Reed Avenue should occur along Snyder Avenue and 17th Street instead of crossing Lincolnway at Reed Avenue. Together, Snyder Avenue, Bent Avenue, 17th Street, and Lincolnway (along with Reed Avenue) are also recommended to be improved with bicycle and pedestrian facilities to connect Pump House Park and the city-wide greenway to the rest of Downtown.



NEEDED PEDESTRIAN AND BICYCLE CONNECTIONS

-  Recommended Reed Avenue Rail Corridor Improvement
-  Recommended Off-Corridor Improvement
-  Existing Bicycle Facilities

THE MARKET AND ECONOMIC OPPORTUNITIES

This market analysis identifies residential and retail development potential and employment opportunities for the West Edge District of Downtown Cheyenne and the Reed Avenue Corridor. A complete version of the market study for the Reed Avenue Corridor is included in the appendix of this report along with three case studies of revitalized warehouse districts in Colorado, Kansas, and Utah.

The City of Cheyenne is the primary market area (PMA) for the West Edge District. Cheyenne is the cultural and economic hub of Southeast Wyoming and has direct economic ties to the highly populated Front Range area of Colorado. Cheyenne is the area from which the West Edge District draws the majority of its residents (housing), patrons (retail), employees (office, institutional), and visitors.

Laramie County, Wyoming, is the secondary market area (SMA) for the West Edge District. It is a much larger area than the PMA, encompassing almost 2,700 square miles. The county boundary mirrors that of the Cheyenne Metropolitan Statistical Area as defined by the U.S. Census Bureau. The SMA borders Nebraska on the east and Colorado on the south. The northern boundary of the SMA is near Chugwater, Wyoming, and its western boundary is near Buford, Wyoming. Additional residents, patrons, employees, and visitors may be drawn from this area, although not as many as from the PMA.

Key Demographic Findings

- ➔ A significant percentage of the West Edge/Downtown population is in the millennial (25 – 34 year) age category.
- ➔ The population of the West Edge/Downtown is slightly more diverse than Cheyenne or Laramie County. All areas are predominantly white, however.
- ➔ Educational attainment is lower in the West Edge/Downtown, although the highest level of education for all three areas (West Edge/Downtown, Cheyenne, Laramie County) is predominantly an associate's degree or some college.
- ➔ Household sizes are smaller in the West Edge/Downtown, with over 60% of households primarily comprised of one person.
- ➔ The median household income in the West Edge/Downtown is significantly lower (\$33,000) than the median household incomes in Cheyenne (\$60,000) and Laramie County (\$63,000). This may be partially attributable to the smaller household sizes. Over 25% of all households in Cheyenne and Laramie County have household incomes of over \$100,000 annually.
- ➔ Most of the residential units in the West Edge/Downtown are rentals, while over 60% of the units in Cheyenne and Laramie County are owner-occupied.
- ➔ Between 2005 and 2015, the jobs with the greatest growth in Laramie County have been in health care and social assistance, followed by transportation and warehousing.
- ➔ While still the majority, the number of persons who both live and work in Cheyenne has declined over the past 10 years.
- ➔ In the past 10 years, there has been a slight increase in older (55+) nonresidents working in Cheyenne and Laramie County and a decrease in younger workers.
- ➔ The percentage of nonresident workers with higher incomes has increased over the past 10 years, perhaps due to age and experience.
- ➔ While there is significant commuting between Cheyenne and surrounding counties, the majority of Laramie County residents (82%) work in Laramie County while 77% of Laramie County workers still reside in Laramie County. The Fort Collins/Northern Colorado connection is present but at a much lower percentage of the working population (5%).



DEVELOPMENT CASE STUDIES

Three redevelopment case studies are presented in their entirety in the appendix of this report: *River North (RiNo) District in Denver, Colorado*; *the Warehouse Arts District in Lawrence, Kansas*; and *the Granary District in Salt Lake City, Utah*. While the case study summaries describe these similar districts' backgrounds in detail, there are several themes that emerged from the studies that are relevant to the revitalization of the West Edge District of Downtown Cheyenne.

- **HEALTHY DOWNTOWN:** All case study examples are located at the edge of their respective downtowns and in each case, the downtowns have seen years of public and private investment which has resulted in vibrant, energetic downtowns which have “spilled over” into the neighboring warehouse districts.
- **NEIGHBORHOOD PLANNING:** The respective jurisdictions have conducted extensive, complementary neighborhood planning, which set visions for future development in each of these districts.
- **SPECIAL DISTRICT FINANCING:** In Lawrence and Salt Lake City, local jurisdictions used redevelopment district tools and tax increment financing to help address infrastructure and financing gaps for specific projects.
- **ARTS AS A CATALYST:** The role of the arts and artists is important. In Denver and Lawrence, artists were among the early pioneers in populating and working in the warehouses. Their roles were later formalized into arts-related organizations and artist supportive activities.
- **PRIVATE-SECTOR CATALYTIC INVESTMENT:** Planning and public investment has been complemented and catalyzed by private-sector investment. Privately funded projects send signals to the market that areas are worthwhile investments. In Denver, before the public sector was even aware of the potential of the area, Mickey Zeppelin with Zeppelin Development developed 25,000 square feet of flexible work space, along with restaurant and other uses. In Lawrence, Tony Krsnich redeveloped the historic Poehler Building into 49 residential units with an affordable housing component which became a catalyst for further development. In Salt Lake City, Artspace developed Artspace Commons, a 102-unit affordable residential project with 50,000 square feet of commercial space comprised of studios, offices, and retail space.



River North (RiNo) District, Denver, CO (Photo Credit: Visit Denver)



The Granary District, Salt Lake City, Utah (Photo Credit: inhibit.com)



Warehouse Arts District, Lawrence, Kansas (Photo Credit: Brent Flanders)

LAND USE RECOMMENDATIONS

One of the challenges in the West Edge District is that there is currently relatively little development activity in the West Edge District to suggest that without public incentives and/or activity, that development/market activity is primed to come to the area.

The following analysis suggests that the West Edge District is a sub-district of Downtown Cheyenne. Albeit different in form and land uses, the West Edge District of Downtown is poised to benefit from the same conditions benefiting downtowns throughout the country, including a desire for a walkable urban environment, locally owned services and amenities, and downtown residential living options.

RESIDENTIAL

Assuming that the West Edge/Downtown is able to capture their current share of future growth, this would result in an estimated demand for 73 units in 10 years for the West Edge/Downtown. There is a property owner in the process of adding some residential units to his primarily commercial (employment) development. If residential does take hold, up to 162 units could be supportable in a decade. One of the market challenges is that there hasn't been a significant increase in urban downtown living in Cheyenne, despite nationwide trends which have seen this happen. The 2015 Cheyenne Laramie County Housing Assessment noted the lack of housing and affordable housing options, as well as a strong desire for downtown living. While there have been units built on the second floors of buildings, significant additions to the building stock haven't taken place. There are an estimated 630 residential units in Downtown currently.



Potential Residential Development Type: Townhomes

RETAIL

Laramie County, like many parts of the U.S., has seen an increase in tax revenue from eating and drinking establishments and a decline in revenue from traditional grocery stores. Retail is increasingly challenged by the increased popularity of Amazon and other retail websites. Nationwide, internet shopping comprises an estimated 10.5% of all revenue from retail and is forecast to continue growing. There has also been a shift towards the retail that is "experiential" rather than focused on acquiring "things." Estimated demand for restaurant and retail uses is estimated at 15,000 to 45,000 square feet over the next ten years.

The initial focus for the Reed Avenue Corridor should be restaurant-related uses. Microbreweries have been mentioned. The current liquor laws, however, appear to be hurdles and difficult for independent restaurants.



Potential "Experiential" Retail Development Type: Craft Brewery

EMPLOYMENT

There haven't been any significant large-scale additions to the office and industrial inventory in the West Edge/Downtown in the last 10 years. While there has been growth, it has taken place primarily in the City of Cheyenne in areas outside of the West Edge/Downtown. Cheyenne (outside of the West Edge/Downtown) has seen an additional 180,000 square feet of office space in the last 10 years, while there has been almost 200,000 square feet of additional industrial space added in the last 10 years.

Employment is forecast to grow at an approximate 1.1% annual growth rate and the West Edge/Downtown still comprises a significant percentage of the current inventory. It is estimated that 25,000 to 100,000 square feet of employment space (office/industrial) could be supportable in the West Edge/Downtown in the next 10 years, although this would likely only materialize if the West Edge reinvents itself into a "hip" place to locate a business.



Potential Employment Development Type: Office/Warehouse

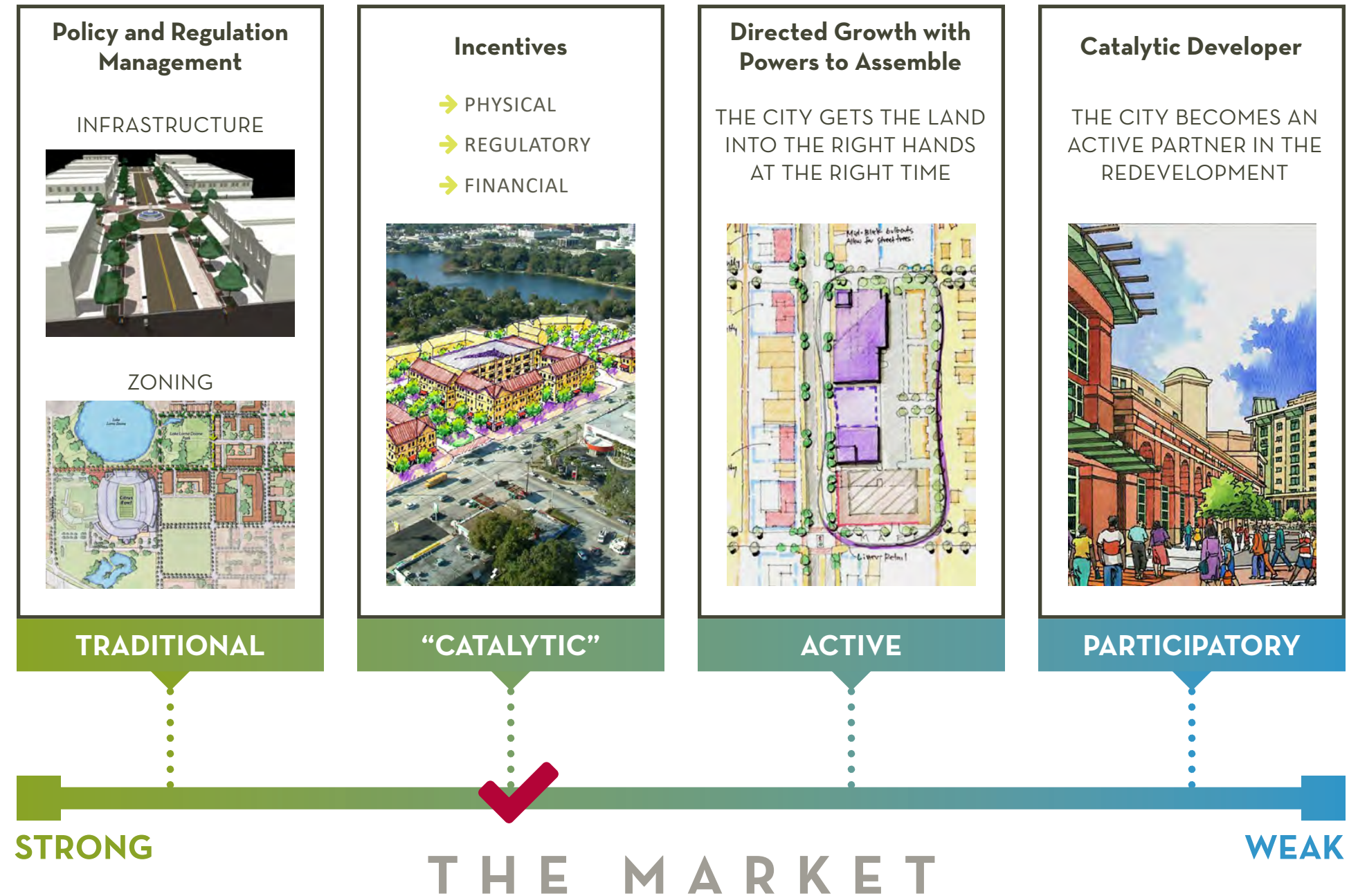
THE CITY AND POTENTIAL CATALYTIC PROJECTS

The idea of catalytic projects was born out of the planning process, from a desire to move forward with revitalization efforts envisioned in prior plans. There are a range of roles the City can play in order to support the redevelopment of the Reed Avenue Corridor. Traditionally, a city's role within a revitalization effort varies based on the strength of the economic market. Based on market conditions, it is recommended that the City play a "catalytic" role, by advancing the redesign of the Reed Avenue Corridor to encourage the private sector to invest in the Corridor.

- **TRADITIONAL ROLE:** At a basic level an "urban design plan" is created with public input and consensus. The plan can then be formalized into land development codes or standards that developers must follow and identify potential public improvements that a city would implement.
- ✓ **"CATALYTIC" ROLE:** Many municipalities utilize incentives, be it physical, regulatory, or financial, to jump start investment in areas struggling economically. These incentives extend beyond traditional municipal infrastructure and regulations to encourage unique investments. The proposed Reed Avenue Rail Corridor Improvement is considered a catalytic investment.
- **ACTIVE ROLE:** Cities can also identify several specific key sites where redevelopment is desired. Using specific redevelopment schemes developed during the urban design process, cities can begin to meet with landowners and potential developers to facilitate relationships that could initiate development opportunities. In this scenario, municipalities typically utilize real estate assets to off-set the cost of development to guide private investment.
- **PARTICIPATORY ROLE:** On the most proactive side is the creation of a public/private catalytic developer. The concept is that through the participation of public and/or not-for-profit institutions or foundations with missions that align with urban redevelopment, and a private developer, projects that would not meet a conventional development pro forma could be made possible through the "public/institutional" financial support. The unique feature of this concept is that both parties are development partners. The private developer gets a return in the short term, the institutional public "developer" gets a return in the long term.

Potential Roles of Cheyenne in Implementing the Reed Avenue Rail Corridor Master Plan

Creating Motivation, Certainty, and Security in the Private Market



CORRIDOR UTILITIES

The Reed Avenue Corridor between Lincolnway and 23rd Street is generally used for three functions: property access, utility conveyance, and railroad connection and service, with the railroad function being the predominant element that is consistent throughout the Corridor. Generally, the Reed Avenue Corridor is free from utility infrastructure. This is likely because of the long-standing easement BNSF owns. **Specific utilities found within the Corridor include:**

Board of Public Utilities

- Water service to the fire station from 19th Street at the east edge of the Corridor
- Water line crossings in the cross streets
- Sanitary sewer crossings at the alley between 21st and 22nd Streets
- Sanitary sewer in Reed Avenue, alley-to-alley on either side of 22nd Street

Black Hills Energy

- 2-inch gas main in Reed Avenue on the east side of the Corridor from 19th Street to 21st Street
- Overhead electric crosses the Corridor at alleys and cross streets
- Potential with cable or telephone sharing the poles
- Overhead electric in a few locations to serve railroad signal cabins

City of Cheyenne

- Storm line crossings in cross streets

The opportunity to provide high-end internet service to the area was a key concern raised during the community discussions. While there are no fiber optics in the Reed Avenue right-of-way, there is abundant space within the Reed Avenue Corridor to establish a utility corridor for both private and City use.

Coordination with fiber optic providers should occur during the design phase to define the amount of conduit capacity; it is expected that four 4-inch conduits would provide opportunities for multiple service providers if the City desires to make an investment in the Corridor.

RAIL CORRIDOR

The Reed Avenue Rail Corridor, from Lincolnway to 23rd Street is generally 80 feet wide and owned by the City of Cheyenne. The current right-of-way and rail corridor is uncontrolled, allowing vehicular and truck traffic to coexist with pedestrian and bicycle traffic all within proximity of the existing rail line.

In the early 1900s, the BNSF railroad was granted an easement to utilize the entire Corridor for their operations. The rail corridor connects with BNSF tracks to the north and the Union Pacific (UP) main line to the south, running generally in the center of the right-of way, shifting slightly east to west as the rail corridor extends north. The City street grid crosses the rail corridor 10 times in the study area.

In addition to the Corridor providing the connection between railroads, rail service has been provided over the years to various industrial customers within the Corridor. Currently, the Tribune Eagle, located between Lincolnway and 17th Streets, is the only active rail service customer in the Corridor. The UP serves the Tribune Eagle. Remnants of past rail service in the Corridor are illustrated by abandoned rail siding in various spots, including roadway crossings in the Corridor. None of these abandoned rail sidings currently connect to the BNSF track.

A detailed inspection of the track within the Corridor to determine the quality of the track materials (ties, rail, fasteners, ballast) was not performed, but there do not appear to be any deficiencies that warrant a decreased level of train operations. BNSF, the State of Wyoming, and the City of Cheyenne recently upgraded all the rail crossings in the Corridor. There is no defined drainage in the Corridor, but it appears that the track is staying dry enough to avoid significant failures to the subgrade, which would cause rapid track structure deterioration.

Given the low use of the track, it likely requires a low amount of maintenance. BNSF has indicated that the track is older and may need more frequent maintenance in the future. Typically, a low-volume track that has been in place for a long time will continue to function quite well and will not be a candidate for significant material upgrades, such as ties, rail, or ballast replacement. For this low-use facility, significant maintenance will only occur if a problem occurs because the railroads typically have higher-priority track maintenance needs elsewhere.



Existing Reed Avenue rail corridor right-of-way



Example of “uncontrolled” access to Reed Avenue



Example of abandoned and disconnected rail siding

FREIGHT RAIL OPERATIONS

BNSF indicates that during its general track maintenance months, April to October, there is approximately one ballast train per day in the Reed Avenue Rail Corridor. It typically arrives on the south from UP and travels north through the Corridor. The Tribune Eagle gets service every few months from UP.

Trains in the Corridor are not controlled by signals, but by line of sight operations. In line of sight operations, trains maneuver at a speed that allows the locomotive engineer to be able to bring the train to a stop for any obstruction within half of his view range down the track, should there be an obstruction such as another train, railroad workers, or a stalled vehicle. Operating speeds for in line of sight operations are 5 mph or less. Streets crossing the track, all at-grade, with railroad crossing signals to control vehicular movements. Signal cabins control the railroad crossing signals and are located within the right-of-way at every road crossing.

Proposed improvements to the Reed Avenue Corridor will not impact or change railroad operations.

Recommended improvements to the Reed Avenue Corridor involve using the unused portions of the right-of-way on either side of the track to provide regional bicycle trail connectivity, similar to Rails-with-Trails improvements that have become common in the U.S. and other countries. Remaining, unused space is planned to be landscaped to beautify the Corridor as well as provide amenity areas for elements such as benches, bike racks, or access to patio/outdoor areas of adjacent parcels as they redevelop.

BNSF requested that operating areas provide a positive drainage for an area 14-foot clear in each direction from the center line of track along with a parallel 14-foot maintenance road. BNSF also suggested Rail operations be separated from pedestrian activity by a 6- to 8-foot fence.

The proposed improved cross section of Reed Avenue is based on defining a safe and acceptable operating corridor for the railroad. The width of the rail corridor was set by providing a minimum operating clearance from the center line of the track to a fixed obstruction, in this case a fence. This minimum clearance varies from state to state, but is in most case 8 – 8.5 feet; for the State of Wyoming, the minimum clearance is 8.5 feet. This clearance provides for a minimum 2-foot walkway at the edge of the envelope for trainman use to throw switches, uncouple cars, etc., and to stand clear of a train on the track. A typical hopper car or a box car will not exceed 11 feet in total width or 5.5 feet on each side of the track center line.

BNSF has requested that the fence at the edge of the envelope be 8 feet high, but has stated that the fence height will be evaluated on a case-by-case basis.

In other mixed-use areas, it is common for the fence to be of a height that deters general trespassing across the railroad track, while also allowing for emergency egress from the rail corridor should someone feel trapped by an oncoming train. This fence height is generally set at 42 – 54 inches.

The proposed fence is 48 inches in height. This fence will prevent individuals that access the Corridor from encroaching the railroad clearance envelope, which is a significant safety upgrade over the Corridor's existing unfenced condition. A good example of a similar treatment to what the project is proposing is the BNSF line through the Convention Center district in San Diego, California.

BNSF has also requested that a parallel maintenance road be provided on one side of the track. A typical maintenance road is 10 feet in width. The project proposes that the adjacent 10-foot bicycle trail can serve that function. A gate is proposed to be installed mid-block at each block, to allow railroad forces doing general maintenance to access the track at the farthest point from either cross street (about 320 feet from edge-of-street to edge-of-street).

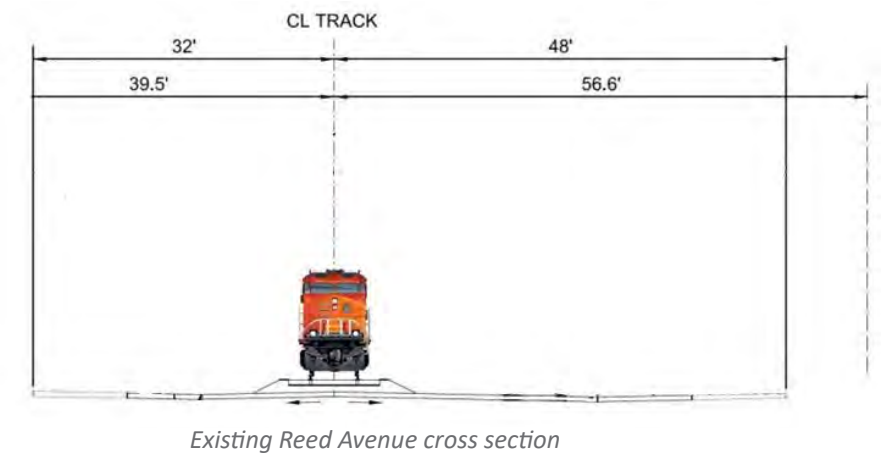
The proposed fence will be designed to be modular so that panels between posts can be easily removed. Given the low amount of train use of the track, there is ample opportunity for the railroad to use track-mounted equipment to access the exact spot that track maintenance would be required.

By formalizing the unused portions of the Corridor, the proposed improvements will be designed to direct stormwater away from the track, collecting it and conveying it to systems in the adjacent cross streets, further extending the life of the track structure and track materials.

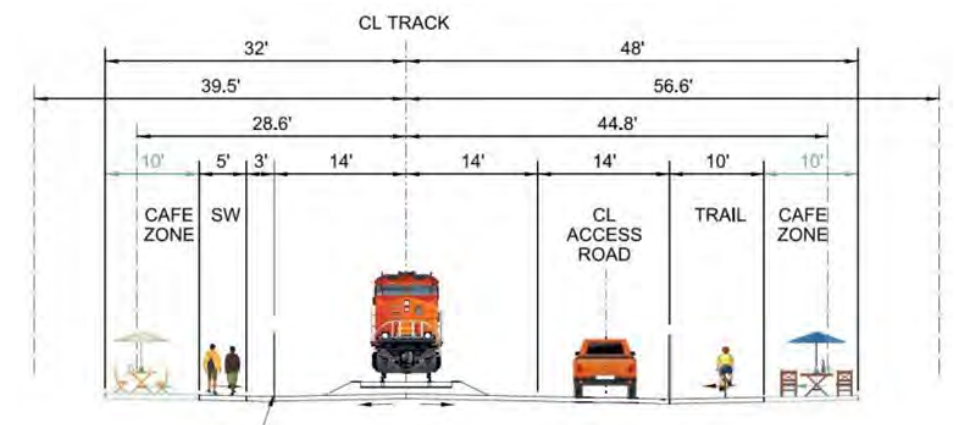
Thoughtful design will incorporate water quality elements such as stormwater planters or bioswales that capture some storm runoff, utilizing the water to irrigate plants and trees while also providing water treatment. Maintenance in the Corridor, outside of the defined track area, will be performed or contracted by the City of Cheyenne, minimizing the railroad's maintenance requirements.

The design will not change how service to the Tribune Eagle is provided, so the railroads will not lose that customer. Signals at road crossings will be maintained in their current condition and may be upgraded in the future to quiet zone levels (see Quiet Zone discussion below).

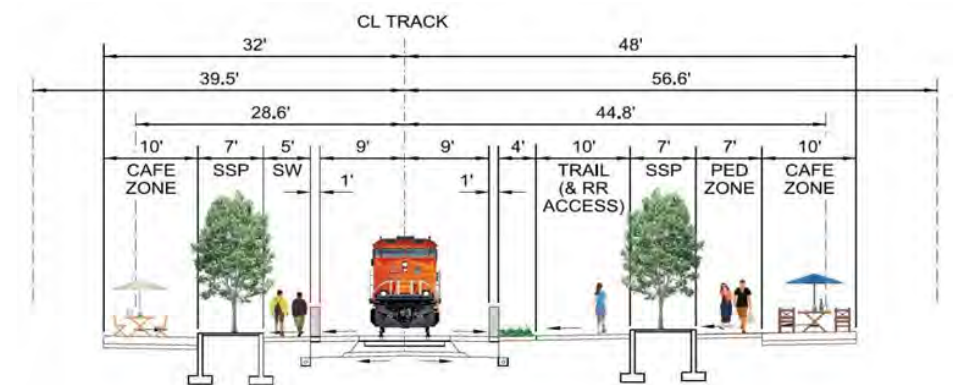
DRAFT



Existing Reed Avenue cross section



BNSF's preferred rail operations cross section requirements



Proposed Reed Avenue cross section

PASSENGER RAIL OPERATIONS

In 2010, the Rocky Mountain Rail Authority (RMRA), a governmental authority made up of over 50 local governmental entities, including the City of Cheyenne, completed the *High-Speed Rail Feasibility Study* (RMRA Study). This effort evaluated the I-25 corridor from Cheyenne, Wyoming to Trinidad, Colorado, passing through the metropolitan areas of Fort Collins, Denver, Colorado Springs, and Pueblo along the way.

The RMRA Study concluded that high-speed transit (HST) is feasible within Federal Railroad Administration (FRA) guidelines on a north-south corridor from Fort Collins to Pueblo. A Cheyenne stop was not considered feasible in this first phase analysis. The most feasible segments and technologies for the Colorado HST were identified for the purpose of ascertaining the most favorable benefit/cost ratio; however, no specific segment or technology was selected or recommended in the study.

The State of Colorado advanced the HST recommendation from the RMRA Study through the 2014 Interregional Connectivity Study (ICS). The objectives of the ICS were to:

- Provide preliminary recommendations for HST segments, technologies, and station locations in the Denver metropolitan area that would maximize ridership for the existing and proposed RTD FasTracks transit system and future HST service.
- Identify potential future HST connections with Denver Regional Transit District (RTD)'s FasTracks system.
- Determine optimal locations for a north-south (Front Range corridor) HST segment from Fort Collins to Pueblo and an east-west HST segment from Denver International Airport to the C-470/I-70 interchange in Jefferson County.

The study concluded with two broad options of advancing HST technologies which terminate in Fort Collins from Denver along either the US-287/BNSF corridor, or I-25. The ICS also concluded that HST along the I-25 corridor and commuter rail (interconnecting cities between Fort Collins and Denver) along the US-287/BNSF corridor could co-exist because they serve different markets. The ICS discussed the potential for commuter technologies to share track between the Longmont interchange area (SH-119/I-25) and RTD's North Metro end-of-line station near 162nd Avenue/Colorado Boulevard.

The *North I-25 Final Environmental Impact Study and Record of Decision* formalized the State of Colorado's position related to both commuter rail and HST. The document officially endorsed commuter rail to terminate in Fort Collins along the US-287/BNSF corridor while HST was identified to terminate in Fort Collins along the I-25 corridor. Options for extending either transit technology north to Cheyenne was not precluded in the state's preferred alternatives.

This Reed Avenue Rail Corridor Master Plan did not evaluate the feasibility of locating either commuter rail or HST technology terminus in the City of Cheyenne. However, this process did review general planning criteria for both technologies for three possible locations within Cheyenne: Reed Avenue, the

Downtown Depot, and Swan Ranch Industrial Park. Of these three possible alternatives, the Reed Avenue Corridor appeared to perform the worst in terms of transportation impacts and land use compatibility. The Downtown Depot performs the best from a land use perspective while Swan Ranch appears to have the fewest transportation impacts among the three evaluated.

QUIET ZONE ASSESSMENT

The FRA requires train operators to sound their horns in advance of every street crossing that does not have adequate warning and vehicular crossing protection equipment in place. Trains are required to sound their horns in a specific sequence and within a specific decibel range for every crossing. The existing close and regular spacing of the 10 crossings in the Reed Avenue Corridor requires trains to be sounding their horns almost continuously until they have passed through the last crossing.

Quiet zones are a program developed by the FRA to eliminate the requirement for trains to sound their horns in advance of every road crossing. A quiet zone is a section of a rail line at least one-half mile in length that contains one or more consecutive public highways or street grade crossings in which locomotive horns are not routinely sounded when trains are approaching the crossings.

The Quiet Zone Program is predicated on the crossings having adequate safety measures which can reasonably prevent a vehicle from entering a railroad crossing when a train is approaching. These safety measures are typically defined as being fully gated or having directional gates with raised medians.

Quiet zones must be developed in accordance with 49CFR222 – Code of Federal Regulations, which establishes a process for local authorities to implement. Each crossing within a proposed quiet zone is evaluated to determine the risk index for a quiet zone crossing against a crossing requiring the horn to be blown. The FRA has an online program that can be used to calculate both risk indexes and determine if the crossing would qualify.

Supplemental Safety Measures (SSM) compliant for quiet zone crossing include:

- Temporary or full-time closure of the public crossing;
- Automatic gates per vehicle traffic direction with median islands or channelizing devices;
- Four quadrant gates (gates that cover the entire crossing on both sides); and
- One-way streets with gates.

Within the Reed Avenue Corridor, only two of the ten existing crossings—the one-way 19th and 20th Street crossings—meet these criteria.

Engineered and non-engineered Alternative Safety Measures (ASM) can also be applied. Engineered ASMs typically are modified SSMs, with some minor deficiency such as median curb type or only three quadrant gates. The local agency must develop engineering justification and obtain FRA approval in order to implement.

Non-engineered ASMs are more passive in nature and include programmed enforcement, public education, and photo enforcement. Because this type of measure does not provide physical restrictions to the track area, many agencies will not consider them as viable solutions to create a quiet zone.

Similarly, railroads may ask for indemnification agreements to protect them. Regular auditing is required to show that the ASMs have been effective and that compliance meets requirements. In the case of either SSMs or ASMs, it is important for a railroad and city to comply with their respective maintenance requirements to minimize liability. Every intersection with a quiet zone will require a separate Construction and Maintenance Agreement between a city and railroad.

In order to develop a quiet zone for Reed Avenue, the Corridor must be at least one-half mile in length. The distance from Lincolnway to 23rd Street would qualify. Installing quiet zone intersection controls would improve the quality of life for residents in and adjacent to the Corridor, but will likely require the City to fund the improvements, as the low usage of the track doesn't present a competitive case to obtain federal funds, i.e., there is a low potential safety risk for rail-to-vehicle conflicts compared to other higher-volume train and vehicle crossings in the country.

It is anticipated that all 10 crossings would require some form of improvement to meet quiet zone requirements. 19th and 20th Streets, though gated one-way streets, would likely still require fiber-optic communication improvements between all of the needed signals. Specific cost estimates for quiet zone improvements were not conducted. Generalized costs could likely exceed \$500k per crossing.

Based on the low number of trains on this seasonal track along with the likelihood of limited or no federal grants, the anticipated costs of quiet zone improvements appear to exceed the benefits. ≡

CONCEPTUAL DESIGN

The Reed Avenue Rail Corridor Master Plan establishes a 30% conceptual design framework for reshaping the Reed Avenue right-of-way between Lincolnway and 23rd Street to accomplish several objectives:

1. Become an urban amenity that it is attractive to a wide variety of land uses, not just industrial activity, and that positions the Corridor for new investment in the West Edge District of Downtown Cheyenne;
2. Create a multimodal corridor which extends the city-wide bicycle and pedestrian greenway from the Ames Avenue underpass and Pump House Park into Downtown Cheyenne; and,
3. Improve the operational safety of the City right-of-way and the BNSF easement.

As stated earlier, a 30% design is conceptual and not final. A 30% design establishes the horizontal, not vertical, controls to the streetscape effort along with estimated costs. The 30% design is the foundation upon which specific design decisions (60%, 90%, and 100%) for Reed Avenue are to be based. More specific design considerations, such as selecting specific tree types, lighting fixtures, and/or paving material, emerge as the project advances through final design.

The conceptual design celebrates Reed Avenue's industrial past and rail history through a contemporary use of concrete, wood, and steel. The proposed design extends a multimodal regional greenway north from Pump House Park to 23rd Street and a pedestrian- and bicycle-only zone adjacent to the rail from 17th Street to 22nd Street. Two segments of Reed Avenue, between Lincolnway and 17th Street and between 22nd and 23rd Streets, are proposed to be shared spaces between vehicles, pedestrians, and bicycles.

The greenway is proposed to extend on the east side of Snyder Avenue and on the south side of 17th Street, from Lincolnway to Reed Avenue. The BNSF rail corridor generally runs down the center of the Reed Avenue right-of-way, shifting east to west. In this design, the greenway extends north one block from 17th Street to 18th Street on the west side of the rail corridor. At 18th Street, the greenway is proposed to shift east and extend north on the east side of the rail line to 23rd Street.

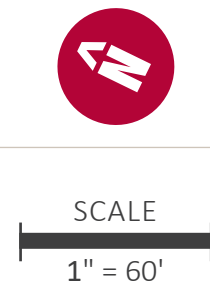
The creative use of alternative hardscape patterns with embedded rail ties reinforce the importance of rail in the Corridor. Finishes, such as lighting, and seating areas will be traditional in their design. The following pages illustrate the design concept developed through this process for the Reed Avenue Rail Corridor.



Illustrative Concept of Reed Avenue between 17th and 18th Streets



NAVIGATE



LINCOLNWAY TO 17TH STREET



Gateway median on Lincolnway



Bike lanes along 17th Street to connect to Downtown



Landscaped area with native grasses



Reed Avenue and 17th Street



Cycle track along 17th Street and Snyder Avenue to connect to Pump House Park

NAVIGATE

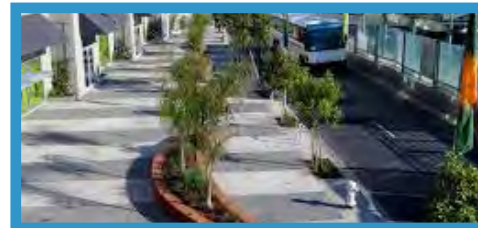


SCALE
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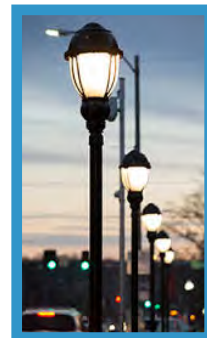
17TH STREET TO 18TH STREET



Integrated art



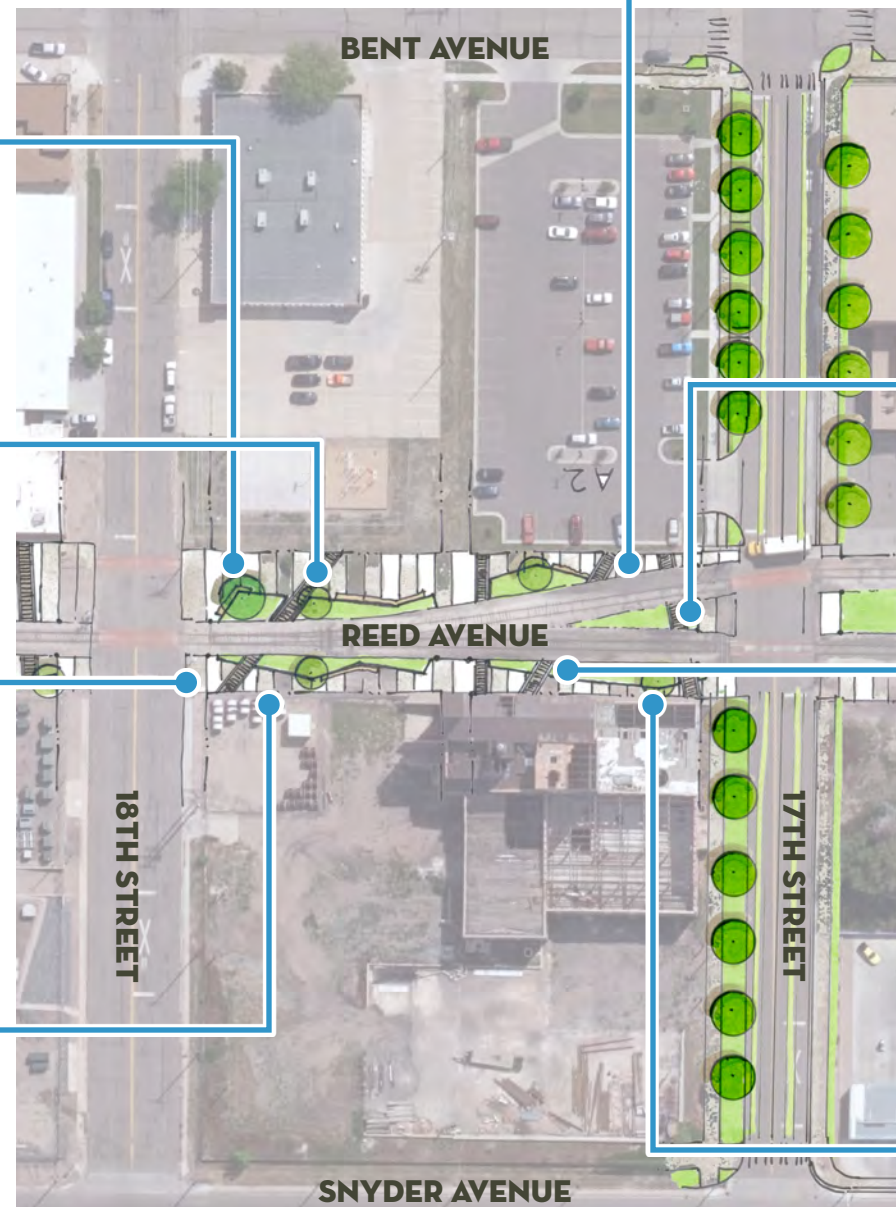
Seating areas



Lighting



Amenities



Looking north from 17th Street



Planting

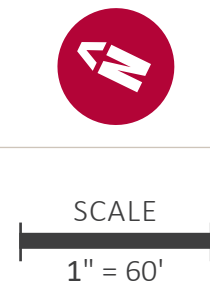


Walls and landscaping



Planting area

NAVIGATE



18TH STREET TO 19TH STREET



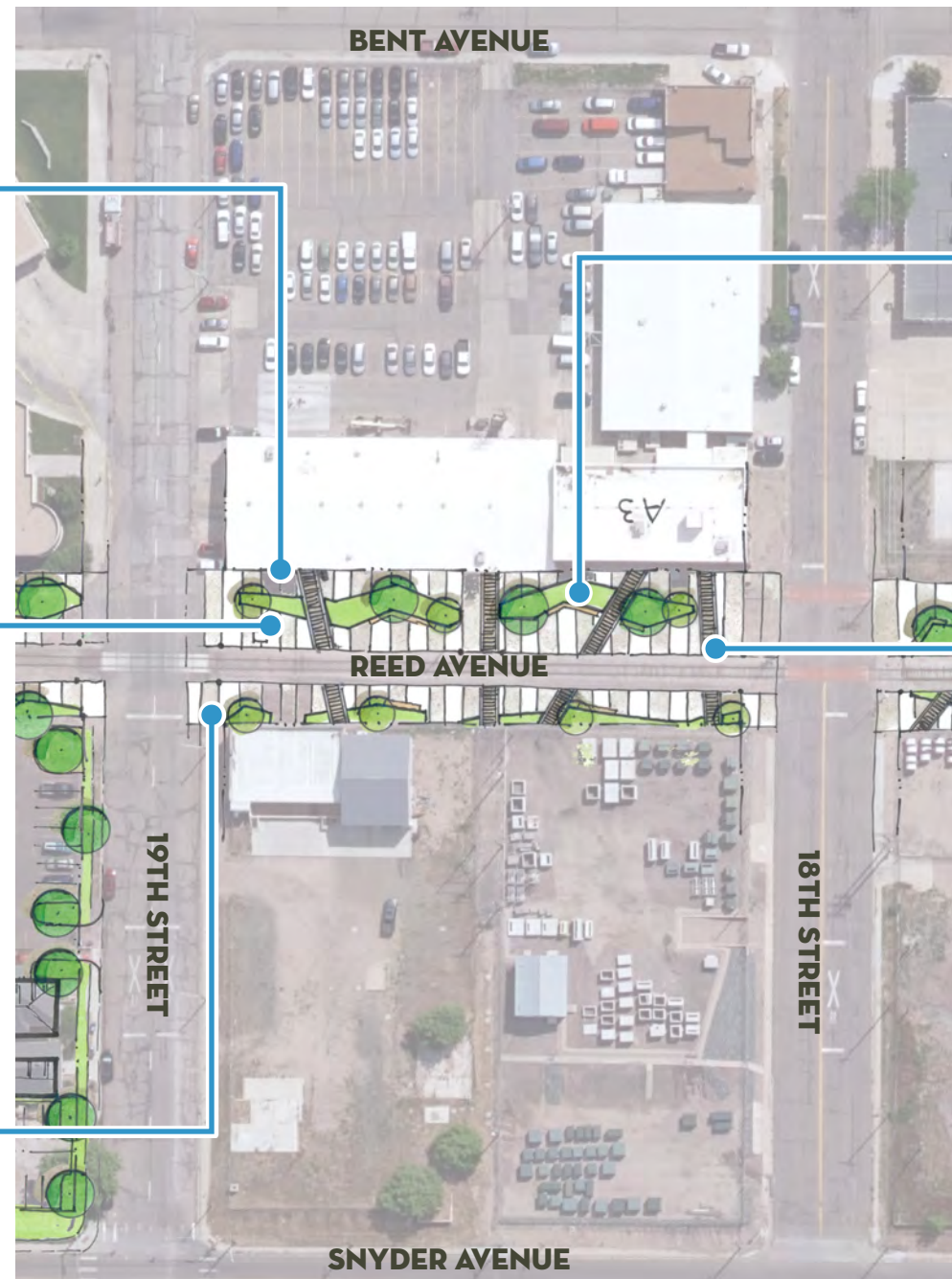
Native planting



Landscape design



Shared streetscape, trail and pedestrian areas

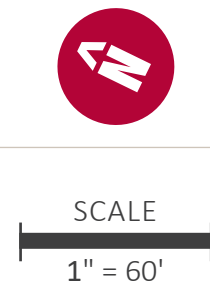


Planting areas near warehouse buildings



48-inch rail fencing

NAVIGATE



19TH STREET TO 20TH STREET



Low maintenance landscaping



Build off of industrial rail heritage

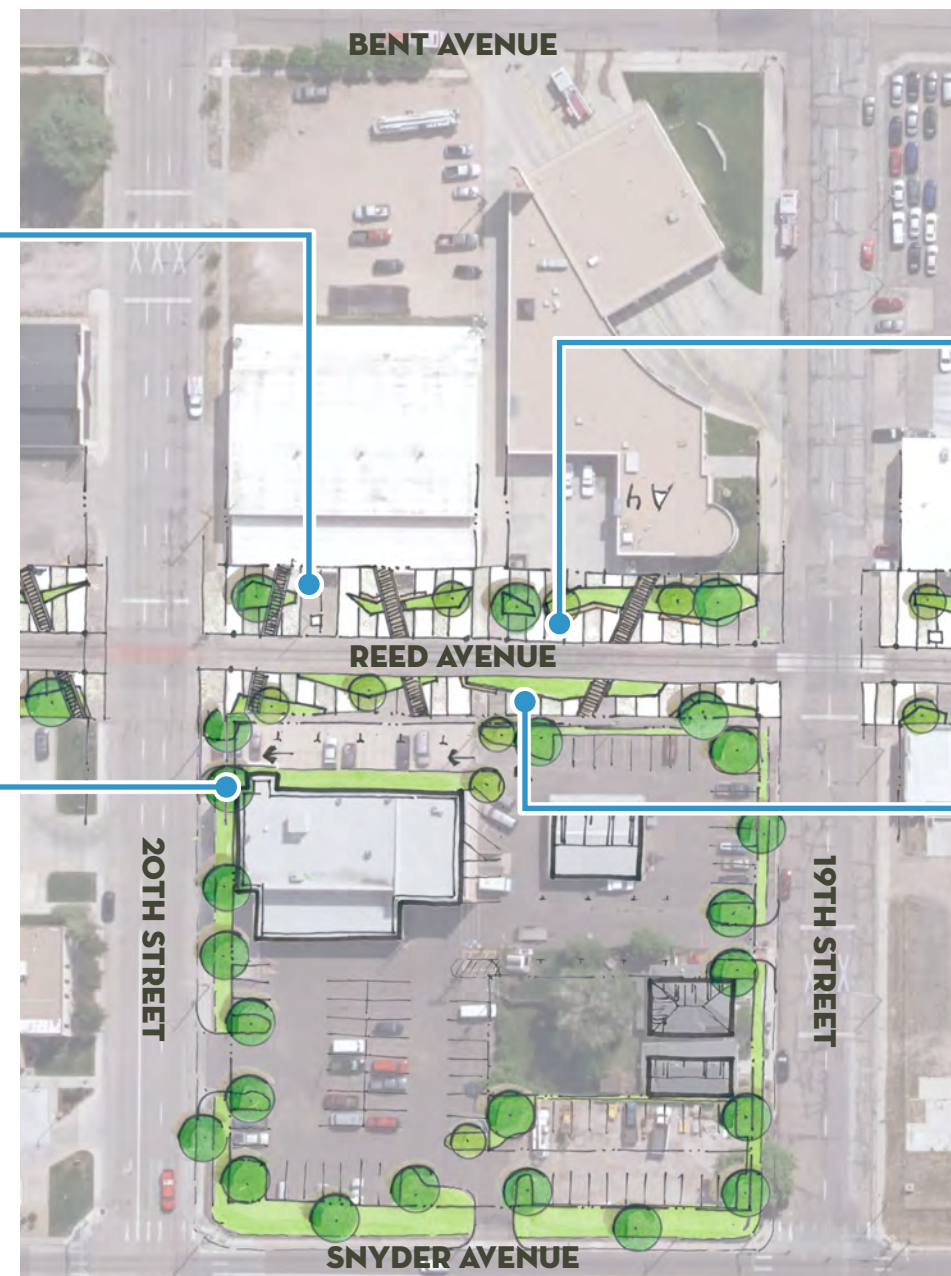
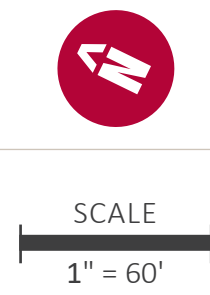


Illustration demonstrating how rail infrastructure can be incorporated into the corridor's redesign



Low-maintenance landscaping

NAVIGATE



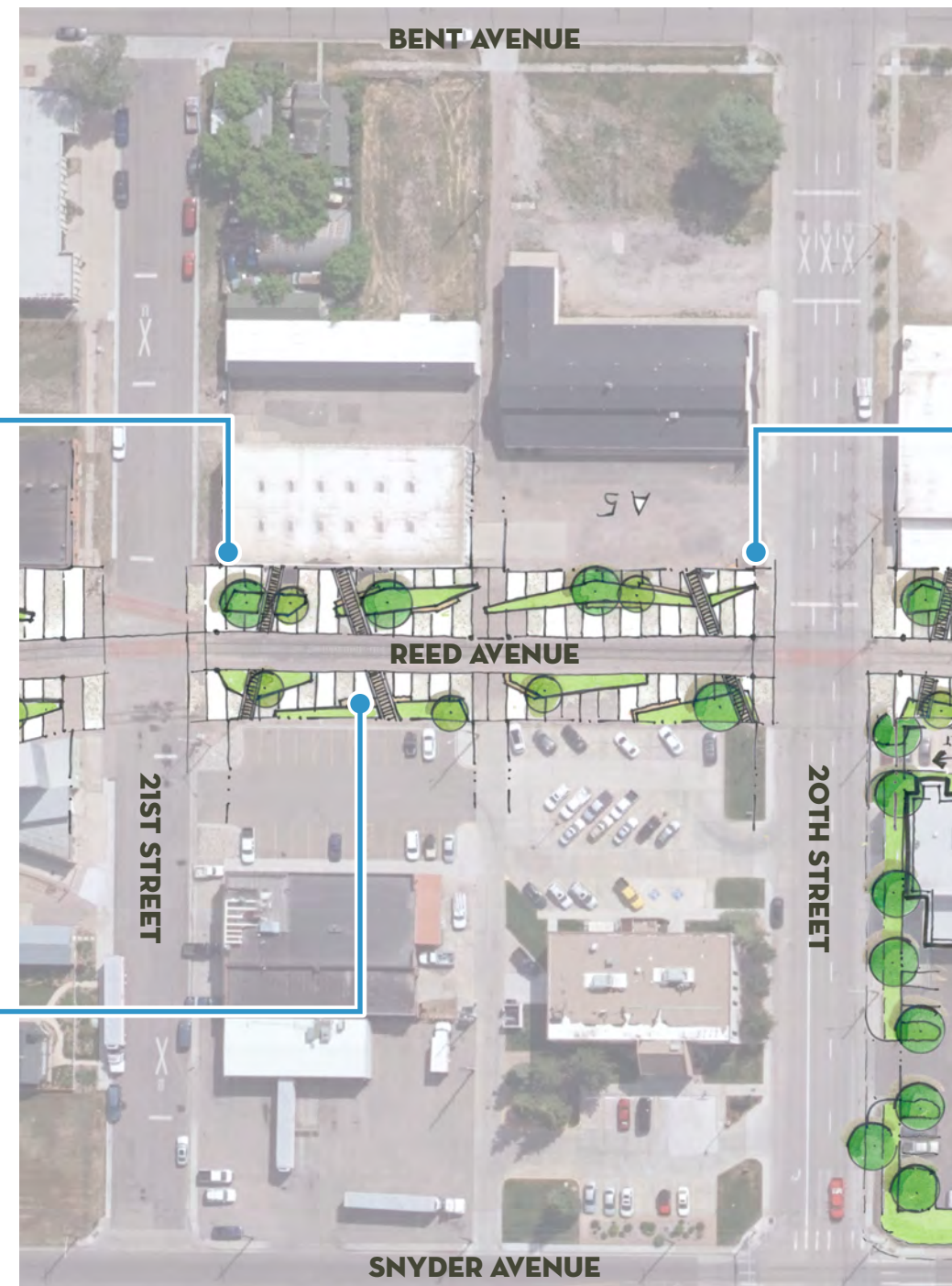
20TH STREET TO 21ST STREET



Low-maintenance landscaping



Integrated seating and landscaping



Trail is separated from pedestrian areas

NAVIGATE



SCALE
1" = 60'

21ST STREET TO 22ND STREET



Integrated seating and landscaping



Large pedestrian area with shade structures



Integrated space with pervious pavers



Seasonal planting arterials and amenities

NAVIGATE

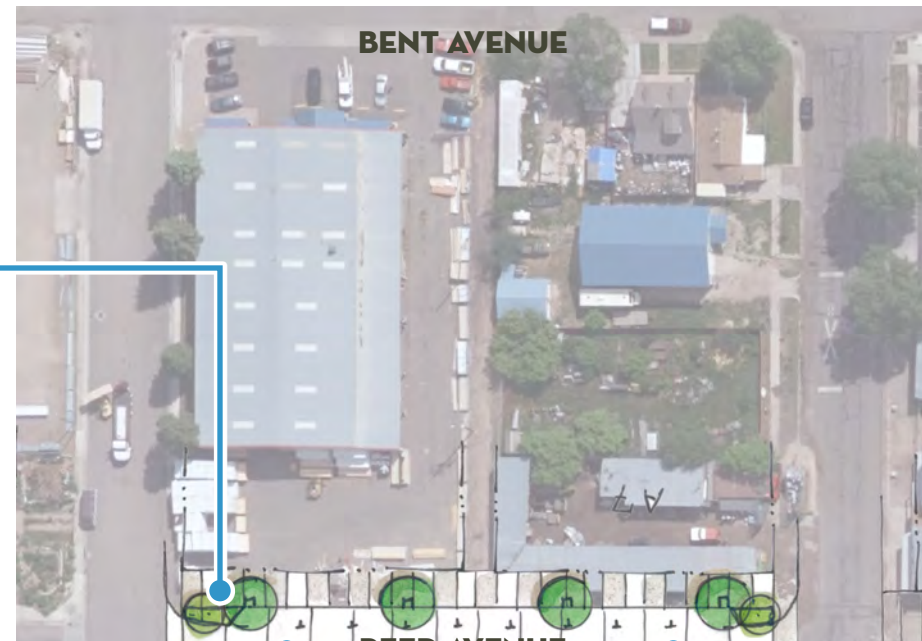


SCALE
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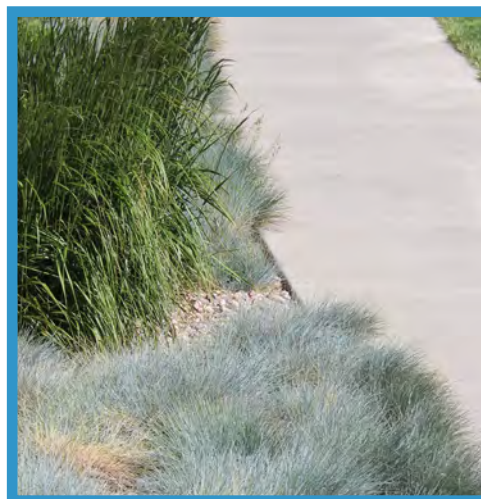
22ND STREET TO 23RD STREET



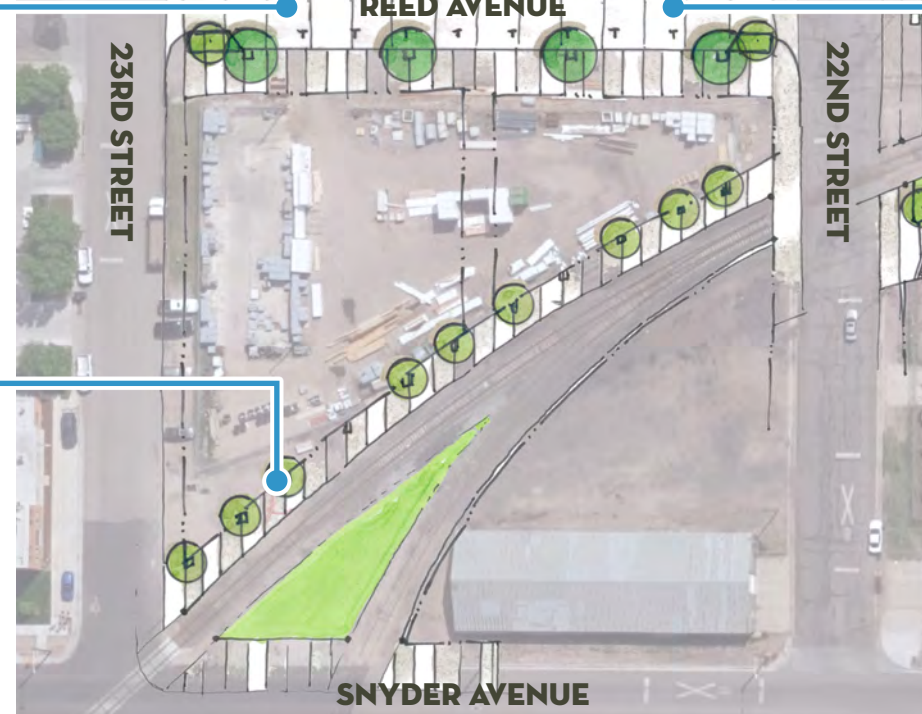
Low-maintenance landscaping



*Alternating hardscape material
is a key design feature in sidewalk*



*Trail continues to follow the rail
corridor to 23rd Street*



*Reed Avenue should function as a shared street,
similar to a European woonerf*



Before



UTILITY MANAGEMENT

Utility potholing of all underground utilities should be completed at the outset of design to define the exact location of all underground utilities. Design elements that impact underground utilities, such as light pole foundations, storm sewers, etc. should attempt to avoid these existing utilities, but relocations may be required. The utility most likely to be impacted will be the Black Hills gas main; although, because this is a relatively small 2-inch line, it may be easy to avoid.

The City may wish to request that overhead electric, and any other services using these poles, be relocated underground to eliminate overhead poles and wires which may eventually be impacted by maturing trees. As design plans are developed showing the location of proposed improvements relative to all utilities, meetings with each impacted utility company should be held to determine if their facility can be protected in place or if it requires relocation.

The opportunity to provide high-end internet service to the area is a key concern. There is abundant space within the Reed Avenue Corridor to establish a utility corridor for both private and City use. This is a relatively inexpensive cost that could provide long-term benefit to the area and redeveloping parcels. While coordination with Century Link and other providers will define the amount of conduit capacity, it is expected that four 4-inch conduits would provide opportunities for multiple service providers if the City desires to make an investment in the Corridor that could potentially be recovered from utility providers in the future.

Alternatively, a utility corridor could be left free of obstructions for future use by a service provider to install within via directional boring or possibly open trenching.

OPINION OF COST

The preliminary cost opinions for the redevelopment of the public infrastructure along Reed Avenue between Lincolnway and 23rd Street were developed using general assumptions and considerations several categories of improvements. Preliminary layouts, derived from a conceptual typical section, were developed for each block of the street by Kimley-Horn in consultation with Wilson & Company.

The project did not collect survey data for the Corridor; therefore, the layouts presented in this chapter are conceptual. Backgrounds for the layouts were developed using aerial photographs and scaled GIS mapping provided by the City of Cheyenne. Information was translated from illustrations developed during the design charrette and converted into CADD drawings shown in the appendix of this report. Quantities, areas, units, and takeoffs used to develop the cost opinion were generated from the electronic drawings.

The categories of improvements considered in the cost opinion furnish all labor, supervision, equipment, tools, material, and other items to complete grading, paving, electrical/lighting, drainage, water, irrigation, fencing, etc. associated with construction.

In all possible cases, values for unit costs were obtained from the *Colorado Department of Transportation (CDOT) 2017 Cost Data Book*. In cases where multiple values were listed for the same improvement, the most conservative average was used. For items which did not have CDOT values available, unit costs were estimated based on Kimley-Horn’s and Wilson & Company’s professional experience in Wyoming and Colorado. This includes knowledge of past projects and research on applicable available products in the Front Range. ≡

SEQ #	DESCRIPTION	UOM	QTY	Unit Cost	Total Cost
Location 1: General					
1	Mobilization/Demobilization	LS	1	\$ 100,000.00	\$ 100,000.00
2	Permit Allowance	LS	1	\$ 50,000.00	\$ 50,000.00
3	Utility Allowance	LS	1	\$ 50,000.00	\$ 50,000.00
4	Traffic Control	LS	1	\$ 50,000.00	\$ 50,000.00
Location 2: Contractor Removal Items					
5	Contaminated Soil Disposal	CY	570	\$ 35.00	\$ 19,945.33
6	Sawcut Pavement	LF	1,120	\$ 3.50	\$ 3,920.00
7	Remove Existing Asphalt Pavement	SY	373	\$ 6.00	\$ 2,240.00
8	Site Cleanup - Remove & dispose of scrap materials, rail, fencing, paving, bollards, etc	LS	1	\$ 20,000.00	\$ 20,000.00
Location 3: Contractor Sitework - Grading					
9	Pollution Prevention Plan - Storm Water (SWPPP)	LS	1	\$ 10,000.00	\$ 10,000.00
10	Excavation - non hazardous waste soils	CY	5,129	\$ 12.50	\$ 64,110.00
11	Subgrade Preparation - 6" Scarify and Recompact	SY	17,096	\$ 3.00	\$ 51,288.00
Location 4: Contractor Sitework - Paving/Striping/Signing					
12	Aggregate - Base Material	CY	-	\$ 70.00	\$ -
13	Asphalt - Patching	LS	1	\$ 20,000.00	\$ 20,000.00
14	Concrete - Unit Paving, Rigid Underlayment	SF	10,900	\$ 10.00	\$ 109,000.00
15	Concrete - Curb and Gutter	LF	868	\$ 25.00	\$ 21,700.00
16	Concrete - 6" Driveway	SY	-	\$ 81.00	\$ -
17	Concrete - 4" Sidewalk	SF	50,000	\$ 7.00	\$ 350,000.00
18	Concrete - 4" Sidewalk (colored)	SY	50,000	\$ 9.00	\$ 450,000.00
19	Concrete - ADA Ramp	EA	14	\$ 2,800.00	\$ 39,200.00
20	Concrete Pavement 8"	SF	20,694	\$ 10.00	\$ 206,940.00
21	Concrete Pavement 8" (colored)	SF	20,694	\$ 12.00	\$ 248,328.00
22	Pavement Marking - Thermoplastic - Symbols/Words	EA	84	\$ 250.00	\$ 21,000.00
23	Pavement Marking - Thermoplastic - Trail Centerline	LF	2,205	\$ 1.80	\$ 3,969.00
Location 5: Contractor Sitework - Miscellaneous					
24	6' Welded Wire Fence with Posts	LF	4,288	\$ 25.00	\$ 107,200.00
25	8' Double Gate, 6' High	EA	6	\$ 1,000.00	\$ 6,000.00
26	Bench, 6'	EA	14	\$ 1,250.00	\$ 17,500.00
27	Trash Receptacle	EA	14	\$ 1,000.00	\$ 14,000.00
28	Bike Racks	EA	7	\$ 1,000.00	\$ 7,000.00
29	Reclaimed Rail	LF	2,956	\$ 25.00	\$ 73,900.00
30	Cast-in-place concrete wall and footing with Skate Protec	LF	661	\$ 70.00	\$ 46,270.00
31	Security Bollard, 34" Height	EA	3	\$ 1,500.00	\$ 4,500.00
32	Security Bollard Footing	SF	102	\$ 4.50	\$ 459.00
Location 6: Contractor Sitework - Site Utilities					
33	18" RCP	LF	2,520	\$ 110.00	\$ 277,200.00
34	Catch Basin	EA	28	\$ 5,500.00	\$ 154,000.00
35	Manhole	EA	14	\$ 4,000.00	\$ 56,000.00
36	Pedestrian Light Fixture, 12'	EA	31	\$ 5,500.00	\$ 170,500.00
37	Site Lighting - Conduit, Wiring, and Controls Complete	LF	2,186	\$ 25.00	\$ 54,650.00
38	Site Electrical - Power service to irrigation, lights, etc	LS	1	\$ 5,000.00	\$ 5,000.00
39	Power Panel/Breaker Box	LS	1	\$ 1,500.00	\$ 1,500.00
40	Potholing	HR	16	\$ 250.00	\$ 4,000.00
Location 7: Landscaping					
41	Intermediate Tree, 6-8'	EA	27	\$ 350.00	\$ 9,450.00
42	Small Tree, 5-6'	EA	50	\$ 250.00	\$ 12,500.00
43	Ornamental Grasses, 1 Gal	EA	4,583	\$ 20.00	\$ 91,660.00
44	Mulch	CY	286	\$ 55.00	\$ 15,730.00
45	Landscape Fabric	SY		\$ 5.00	\$ -
46	Tree Grates	EA	17	\$ 1,500.00	\$ 25,500.00
47	Amended Planting Soil	CY	395	\$ 75.00	\$ 29,625.00
48	Irrigation System - Water Lines, Controller, Wiring, Heads, Emitters, etc	SF	30,985	\$ 1.50	\$ 46,477.50
				Subtotal Construction Items	\$ 3,122,261.83
Construction Contingency				25%	\$ 780,565.46
				Total Construction Items	\$ 3,902,827.29
Railroad Flagging				Day 90	\$ 1,200.00 \$ 108,000.00
Engineering Fee				10%	\$ 390,282.73
Construction Management				17%	\$ 663,480.64
				Total Design and Construction Cost	\$ 5,064,590.66

IMPLEMENTATION ACTION PLAN

The implementation strategy for the Reed Avenue Corridor Master Plan centers on the project's anticipated fiscal impacts and financial benefits. This concluding section of the report includes a summary of the project's fiscal impact assessment and outlines recommended actions for railroad coordination and project funding.

FISCAL IMPACT

Community investment in the Reed Avenue Corridor presents the City of Cheyenne with an opportunity to catalyze the redevelopment of the West Edge District of Downtown as envisioned in *Historic West Edge Reimagined* (2014) and the subsequent *West Edge Area Wide Plan* (2016).

The fiscal assessment evaluated potential tax increments over a 25-year time horizon for all increments, both property and sales taxes, along with Cheyenne's portion of property and sales taxes alone. The analysis assumed that 20 mills for commercial projects are levied on properties currently within the Cheyenne Downtown Development Authority (DDA) boundaries.

The financial/fiscal analysis assumed area improvements in Reed Avenue would result in economic development projects consistent with the market forecast for the West Edge District presented in Section Three of this report.

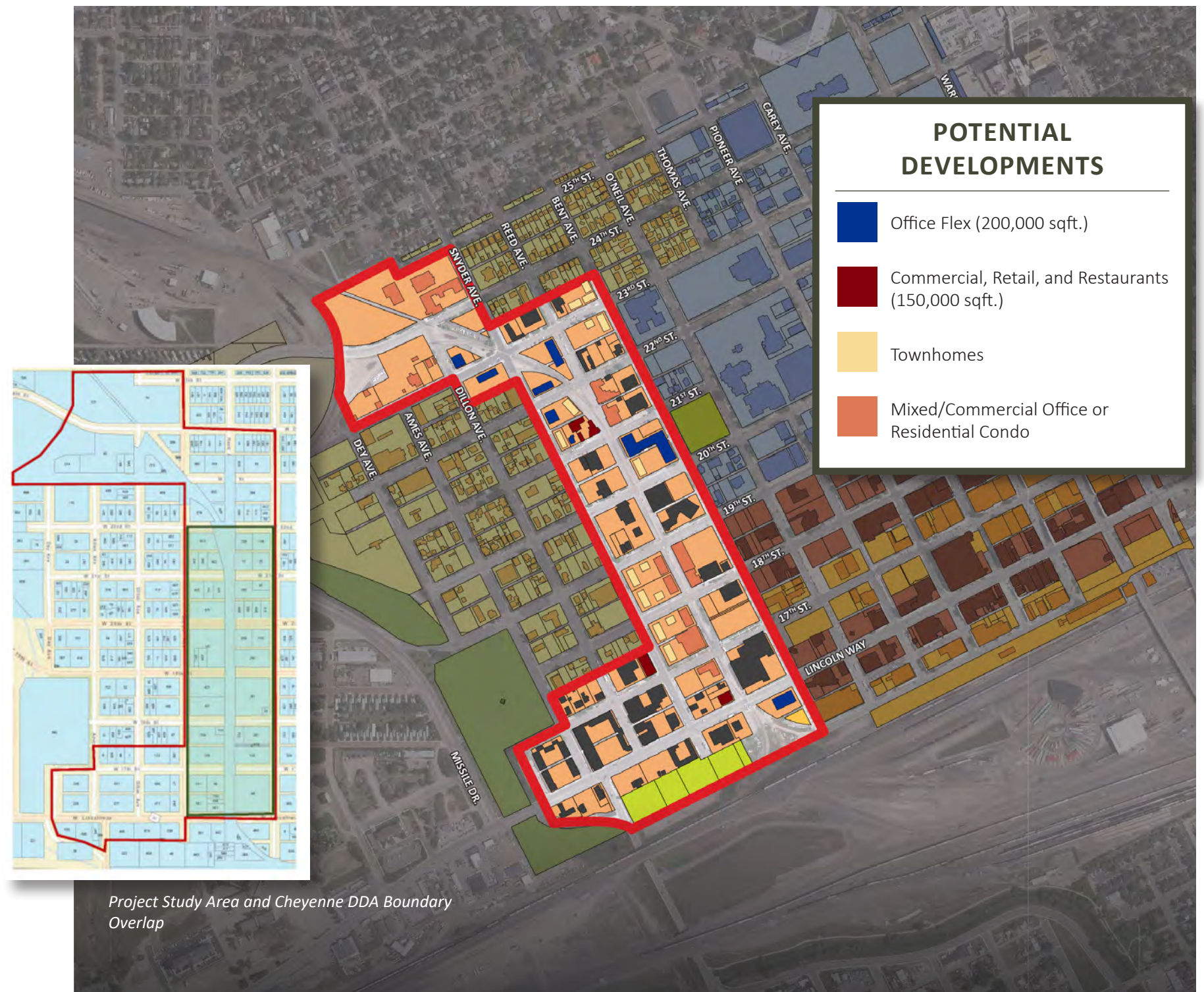
The anticipated development included maintaining current industrial land uses along with additional office flex, commercial/retail/restaurants, townhome, and mixed uses.

The anticipated additional development mix included:

- 180,000 square feet of office/flex/employment uses
- 68 townhomes
- 35,000 square feet of commercial/retail/restaurant uses
- 110,000 square feet of "mixed uses," which could include a combination of employment and residential uses

Property development values were assumed as follows:

- Townhomes valued at \$190,000 per unit
- Office/flex valued at \$100 per square foot
- Commercial and mixed valued at \$150 per square foot



Project Study Area and Cheyenne DDA Boundary
Overlap

Illustrative rendering of a potential development scenario which accommodates the West Edge District's market forecast.

The overall fiscal impact of the anticipated redevelopment assumed that everyone in the community benefits from the property taxes (at 72.2 mills plus 20 mills for commercial DDA properties) and the sales taxes (4% state/local and 2% local) that are levied. The benefits include an increase in property tax collections as well as property and sales taxes for new commercial projects in the area. They are outlined in **Table 1**.

TABLE 1: FISCAL IMPACTS	
CURRENT	
Current Annual Property Tax Generated	\$203,000
NEW PROPERTY TAX GENERATION [1][2]	
Annual Property Tax	\$203,000 increasing to \$943,000 annually
Cumulative Property Tax (25 years)	\$15.2 million
NEW SALES TAX GENERATION [2][3]	
New Sales Generation	\$16 million to \$20 million annually
New Sales Tax Generation (state and local)	\$1 to \$1.2 million annually
Local Sales Tax Generation	\$370,000 to \$450,000 annually with cumulative total of \$4 million
New Jobs	540

- 1) Assumes that property redevelopment commences in 2022.
- 2) Assumes a 2% average annual increase in property and sales values from 2017.
- 3) Assumes sales are generated at approximately \$350 per square foot with 35,000 square feet of commercial space. Development starts in 2032.
- 4) Sales tax generation (4% state/local and 2% local only).

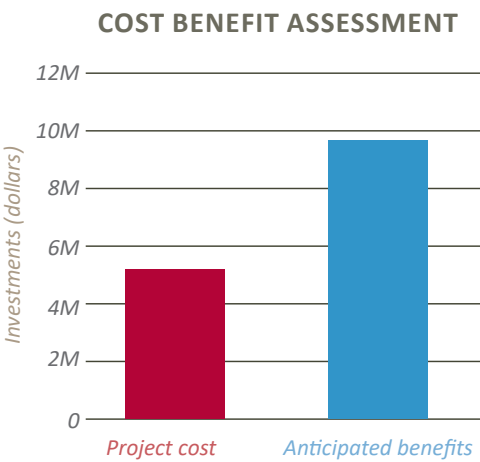
Because the Reed Avenue Corridor currently generates some property taxes, **Table 2** summarizes the anticipated “net new” property and sales tax generated. It assumes that 35,000 square feet of commercial (sales estimated at \$350 per square foot) generates new sales tax revenues for the area.

The total in **Table 3** shows just the calculation based on Cheyenne’s portion of property taxes (8 mills) and its portion of sales taxes (Cheyenne receives 68.37% of 31% of the 4% statewide sales tax. Cheyenne also receives 68.37% of a 2% local sales tax).

TABLE 2: TAX INCREMENT (TOTAL)	
CURRENT	
Current Annual Property Tax Generated	\$203,000
NEW PROPERTY TAX GENERATION [1][2]	
Annual Property Tax Increment	\$4,000 to \$740,000 annually
Net New Cumulative Property Tax	\$9.7 million
NEW SALES TAX GENERATION [2][3]	
New Sales Generation	\$16 million to \$20 million annually
New Sales Tax Generation (state and local)	\$1 to \$1.2 million annually

TABLE 3: TAX INCREMENT (CHEYENNE ONLY)	
CURRENT	
Current Annual Property Tax Generated	\$46,000
NEW PROPERTY TAX GENERATION [1][2]	
Annual Property Tax Increment	\$930 to \$85,000
Net New Cumulative Property Tax	\$1.0 million
NEW SALES TAX GENERATION [2][3]	
New Sales Generation	\$16 million to \$20 million annually
Local Sales Tax Generation	\$370,000 to \$450,000 annually with cumulative total of \$4 million
New Jobs	540

It appears the \$5.1 million investment in the Reed Avenue right-of-way could yield a cumulative \$9.6 million property tax increment within the West Edge District.



RAILROAD COORDINATION PLAN

Initial discussions have been held with BNSF and they are aware of the City’s intent to reinvigorate the Corridor via the implementation of public amenities in the Reed Avenue right-of-way. BNSF’s initial position is that they do not want trails or public accesses running parallel to their track. There are precedents in other locations for implementing this type of improvement. It will be important to stress that the improvements better define their limited-use corridor and prevent trespassing, which could interfere with their operations. The Corridor will have higher public activity, which should help reduce the potential for theft and vandalism.

BNSF will follow a formal process for reviewing any proposed changes. The first step to communicating the project officially to BNSF will be submitting a plan with enough detail to define the area they will operate in and how it affords them to continue operations and maintenance of their track and signal bungalows.

BNSF will review the plans and provide comments, typically hiring a third-party consultant to review the plans. There will likely be several rounds of submittals from concept or preliminary design through final design. The City should also request the retirement of disconnected track facilities that are present in the Corridor. This is a formal process that allows the railroad to determine if the facilities are needed, and, if not, have them removed. BNSF may allow or require the City to remove the track materials. Retiring these elements frees the railroad of having to maintain extra roadway crossings or be responsible for existing remnant tracks. This will further narrow the railroad’s limits of liability and reduce long term maintenance and replacement costs.

A Construction and Maintenance Agreement will be developed that defines both the City’s and BNSF’s obligations for building and maintaining improvements in the Corridor. A final item will involve the current easement agreement between BNSF and the City. BNSF recognizes that the current easement language is old and needs to be updated to reflect current or changed needs and modernize the contract language. Elements defined in the Construction and Maintenance Agreement will be incorporated. It is likely that BNSF will request to extend their agreement further into the future than the existing agreement. The process to obtaining agreement in the Corridor will involve reconciling each of these elements, and each has some impact on the other.

Steps

1. Refine design concept to a point that the City is comfortable with presenting it to the railroad(s).
2. Request railroad agreements with third-party entities.
3. Negotiate design modifications with the railroad(s) and any third parties.
4. Review design with public utilities commission (PUC).

No Action – by Wyoming Statute this is most likely

Action

Submittal for review
- 29

- ii. Field diagnostic reviews for crossings
- iii. Public hearing
- 5. Request retirements of unused track and other railroad equipment.
- 6. Complete final design process with requisite BNSF reviews for final railroad approval.
- 7. Renegotiate the easement agreement.

PROJECT FUNDING AND NEXT STEPS

On May 2, 2017, the citizens of Cheyenne voted to extend a 6th-cent sales tax (1%) to fund several infrastructure improvements throughout the City. Nine propositions incorporating a wide variety of initiatives were included on the ballot. Two of the nine propositions included funding that is either directly related to, or could be considered a part of, the Reed Avenue Rail Corridor Master Plan.

Proposition 9 identified funding directly related to the Reed Avenue Corridor. Specifically, Proposition 9's approval allocated \$4 million "to the City of Cheyenne's West Edge District to construct infrastructure and enhancements and apply for additional grant monies supporting redevelopment, including the Reed Avenue Rail Corridor."

Interestingly, monies from the also-approved Proposition 5 could be considered in partnership with implementing the Reed Avenue Rail Corridor Master Plan. Proposition 5's approval allocates \$3.8 million "to the City of Cheyenne for planning, design and construction of the Greater Cheyenne Greenway." Two of the projects identified for the Greater Cheyenne Greenway in Ward 1 of the City have direct links to the Reed Avenue Corridor Plan. These are:

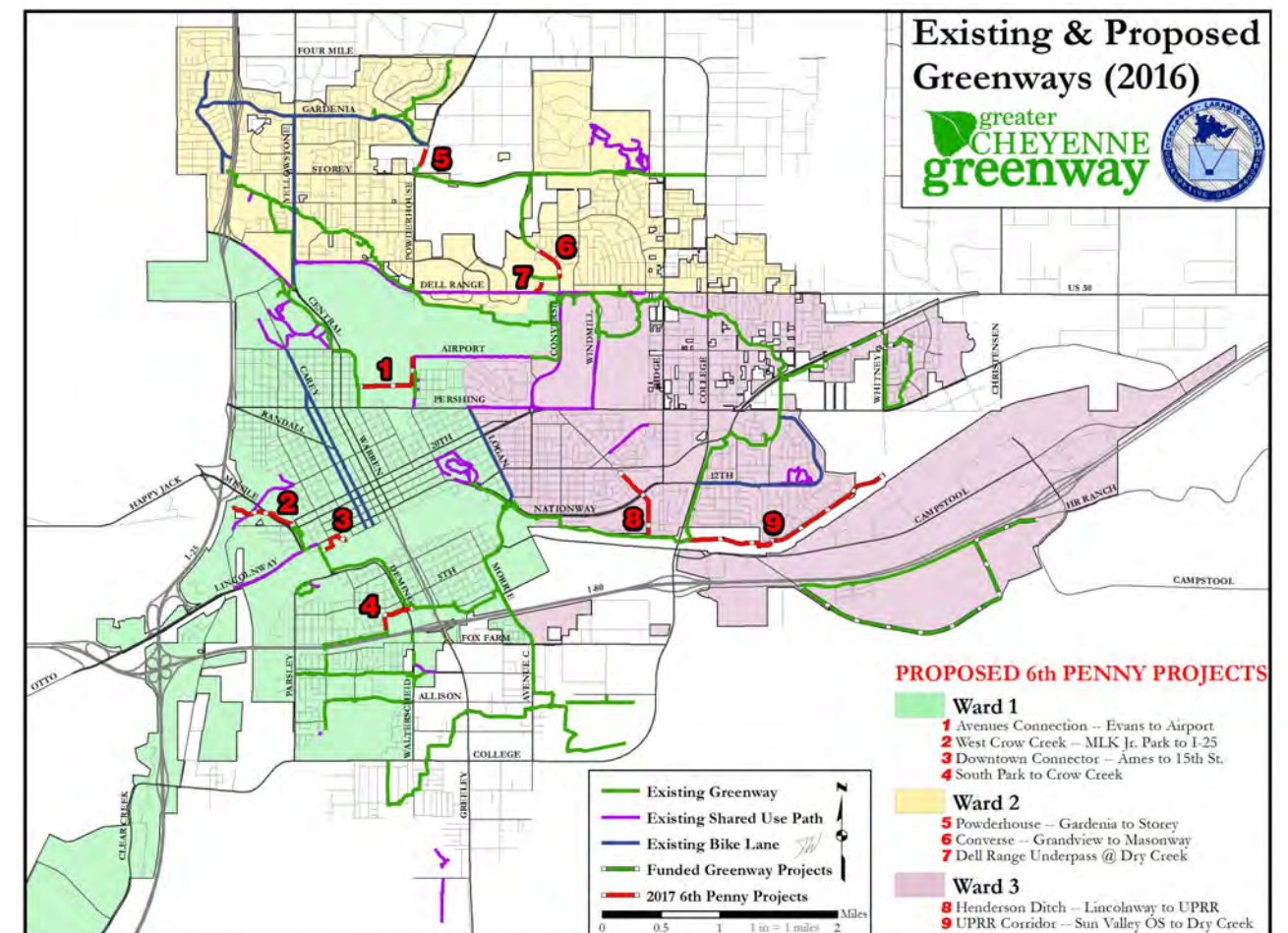
- The West Crow Creek – Martin Luther King Jr. Park to I-25; and
- Downtown Connector – Ames Avenue to 15th Street

The project's initial \$5.1 million opinion of cost will be modified as the project advances through the 60%, 90%, and 100% final design phases. During the final design effort of the project, the community will be presented with choices on important decisions that will impact the final costs of the project. If the community only has \$4 million available through the Proposition 9 funds, the project's design can be scaled back through final choices in paving materials, hardware, and landscaping. However, it is the recommendation of this report for the City of Cheyenne to leverage the Proposition 9 funding "with additional grant monies" as allowed in Proposition 9 to implement the design concepts envisioned in this Master Plan.

Potential funding beyond Proposition 9 funding could come from Cheyenne's DDA. Currently, 60% of the Reed Avenue Corridor falls within the boundaries of the DDA. It is the recommendation of this report for the City to work with the DDA on expanding its boundary to include the entire Reed Avenue Corridor.

Additionally, the City or DDA could pursue additional grant funding as identified by Ayres and Associates, October 1, 2016. Sources include (as identified):

- Transportation Investment Generating Economic Recovery (TIGER): USDOT
- Economic Development Administration (EDA) Planning Program and Local Technical Assistance
 - U.S. Department of Commerce
 - Cool & Connected: Environmental Protection Agency (EPA) and U.S. Department of Agriculture (USDA) Program
 - EDA Public Works and Economic Adjustment Assistance: U.S. Department of Commerce
 - Community Development Block Grant
- Cheyenne-Laramie County Economic Development Joint Powers Board
- Wyoming Department of Transportation (WYDOT) Transportation Alternatives: WYDOT
- Wyoming Business Council
 - Business Committed Grant
 - Community Readiness Projects
 - Community Enhancement Grant
 - Business Ready Community Planning Projects



DRAFT

REED AVENUE
RAIL CORRIDOR
MASTER PLAN

APPENDIX A

MARKET & ECONOMIC OPPORTUNITIES MEMO

2.2 Population and Household Characteristics

The analysis compares the West Edge / Downtown to the City of Cheyenne (PMA) and Laramie County (the Secondary Market Area). Our analysis combines the West Edge and Downtown in order to get a better demographic understanding of who is living in or close to downtown Cheyenne.

Population and Households

Cheyenne’s current population estimate of just over 64,000 represents about 65% of Laramie County’s population, and about 11% of the state’s population. With about 1,340 residents, the population of the West Edge/Downtown comprises about 2% of the city’s population. The population of the West Edge/Downtown has been essentially the same since 2010, with a compound annual growth rate of about 0.5%, about half that of Cheyenne and Laramie County over the same period.

Table 1
Population and Households, 2016-2017

	West Edge/Downtown [1]	City of Cheyenne [2]	Laramie County [2]
Persons	1,338	64,019	98,136
Households	639	27,989	41,256
Persons / Households	2.09	2.29	2.38

Source: American Community Survey, US Census, Claritas, ArLand

[1] 2017 estimates from Claritas

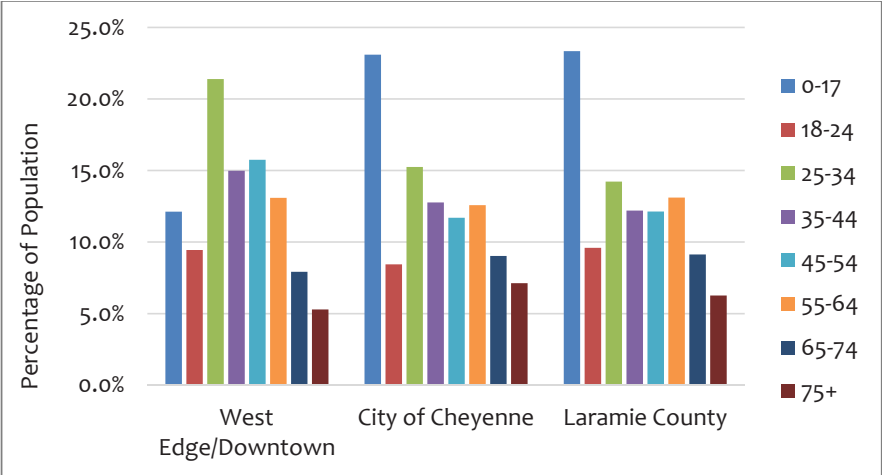
[2] 2016 population estimates from American Community Survey;

household estimates from 2011-2015 American Community Survey 5 Year Estimates

Age Distribution

The age distribution of the West Edge/Downtown differs from that of Cheyenne and Laramie County. The median age in the West Edge/Downtown is higher than the city and county, partially because the share of the population aged seventeen or under (12.1%) is about half that of the city and county. Over 21% of the West Edge/Downtown population is a Millennial (aged 25-34), a much higher percentage than Cheyenne (15%) and Laramie County (14%). Almost 44% of the West Edge population is between the ages of 35 and 64. About 37% of the Cheyenne and Laramie County population is in this age group.

Figure 2
Population by Age 2017



Source: Claritas, ArLand

Race/Ethnicity

The West Edge/Downtown has a lower percentage of the population identifying as white, and a higher percentage of the population identifying as Hispanic or Latino, than the city and county. In addition, about 7% of the West Edge/Downtown population identifies as being black, which is higher than the city and county. This is also true for those identifying as an American Indian or Alaska Native.

Table 2
Race / Ethnicity, 2017

Race/Ethnicity	West Edge/Downtown	City of Cheyenne	Laramie County
White	80.3%	85.1%	86.0%
Black	7.0%	4.3%	3.7%
American Indian	2.3%	1.1%	1.1%
Asian/Pacific Islander	1.3%	1.5%	1.4%
Other / Two or more	9.1%	8.0%	7.8%
Hispanic or Latino*	17.7%	16.0%	15.1%

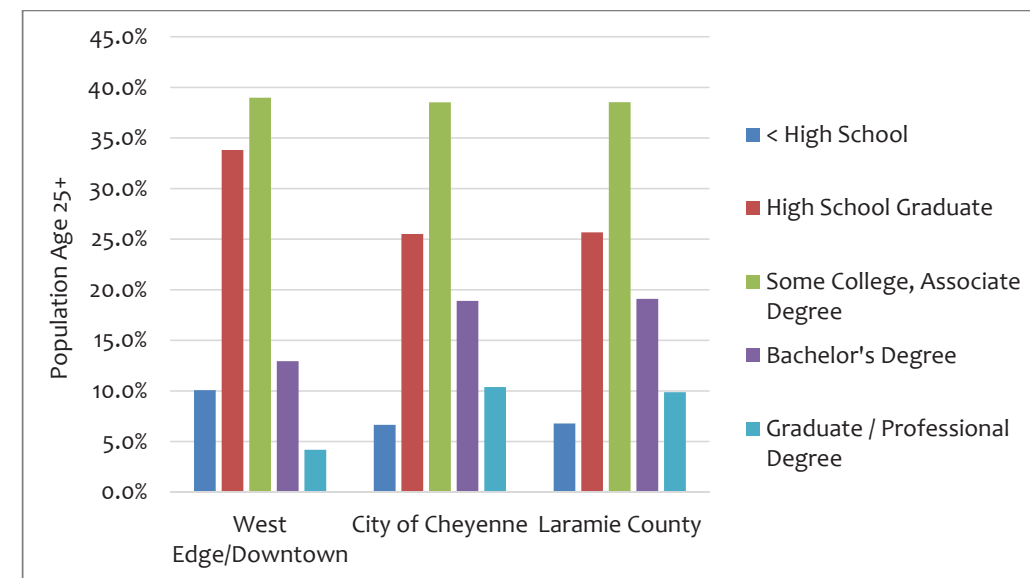
Source: Claritas, ArLand

* can be of any race

Educational Attainment

Educational attainment for the West Edge/Downtown is lower than the City and County. A higher percentage of the population has less than a high school diploma while a lower percentage of the population has a bachelor's degree or higher. About 17% of the West Edge/Downtown population has a bachelor's degree or higher, lower than the city and county at approximately 29%.

Figure 3
Educational Attainment 2017

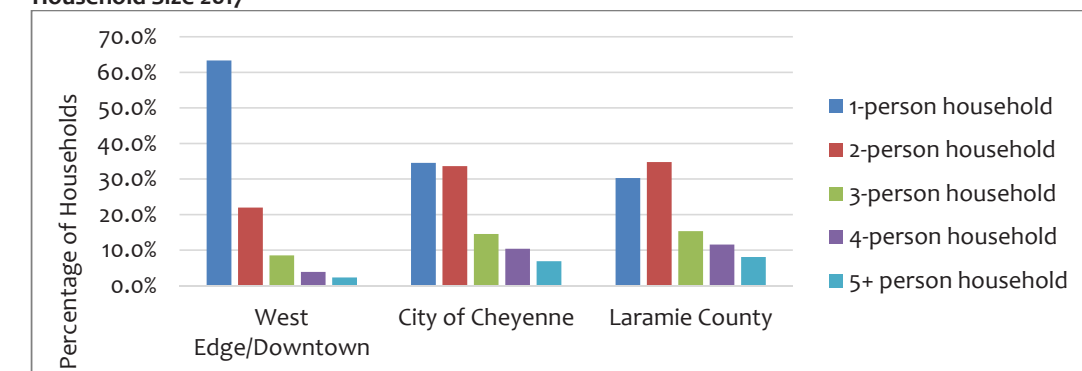


Source: Claritas, ArLand

Household Size

The average estimated household size of the West Edge/Downtown is 1.6 persons, lower than the city and county at 2.3 and 2.4 persons respectively. As seen in Figure 4, about 63% of households in the West Edge/Downtown are 1-person households, more than twice that of the county and almost twice that of the city. Less than 15% of households in the West Edge/Downtown have three or more persons in them.

Figure 4
Household Size 2017



Source: Claritas, ArLand

Household Income

Household income in the West Edge/Downtown is significantly lower than in the city and county. The median household income of the West Edge/Downtown, as seen in Table 3, is about half that of the city and county. Households in Laramie County have average incomes more than \$32,000 higher than those in the West Edge/Downtown.

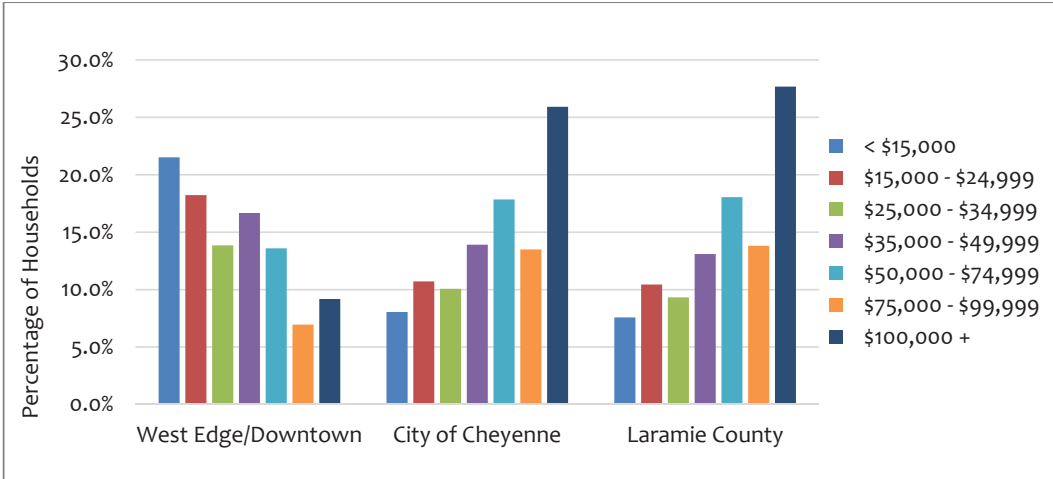
Table 3
Median and Average Household Income 2017

	Median Household Income	Average Household Income
West Edge/Downtown	\$32,398	\$44,850
City of Cheyenne	\$60,185	\$74,048
Laramie County	\$63,233	\$77,202

Source: Claritas, ArLand

As seen in Figure 5, the income distribution is significantly different for the West Edge/Downtown compared to the city and county. 21.5% of households in the West Edge/Downtown have incomes of less than \$15,000. Only about 8% of the city and county have incomes in the same income bracket. About 30% of households in the West Edge/Downtown make \$50,000 or more. In the city and county, this figure is about 47% and 60%, respectively.

Figure 5
Household Income 2017



Source: Claritas, ArLand

Housing Tenure

Rental housing makes up about 72% of all occupied housing units in the West Edge/Downtown unlike the city and county where owner occupied housing is the majority. Rental units make up about 36% and 32% in the city and county, respectively.

Table 4
Occupied Housing Units by Tenure 2017

	Owner Occupied	Renter Occupied
West Edge/Downtown	27.6%	72.4%
City of Cheyenne	64.4%	35.6%
Laramie County	68.4%	31.6%

Source: Claritas, ArLand

2.3 Employment

Average annual employment in Laramie County has increased by about 4,600 jobs over the ten-year period from 2005 to 2015. The largest jobs increase was in the Transportation and Warehousing Sector, followed by Health Care and Social Assistance. However, in percentage terms, the Mining Sector, which includes oil and gas extraction, saw the largest compound annual growth of almost 20% during this time. The drop in oil prices that began in the summer of 2014 undoubtedly has had an impact on recent employment in this sector.

Table 5
Laramie County Annual Average Employment by Industry, 2005-2015

	2005	2010	2015	Change (2005-2015)	CAGR [1] (2005-2015)
Accommodation and Food Services	4,173	4,004	4,532	359	0.8%
Administrative and Waste Services	1,497	1,638	1,637	140	0.9%
Agriculture, Forestry, Fishing & Hunting	170	279	341	171	7.2%
Arts, Entertainment, and Recreation	272	317	371	99	3.2%
Construction	2,765	2,621	3,354	589	1.9%
Educational Services	140	226	241	101	5.6%
Finance and Insurance	1,501	1,647	1,700	199	1.3%
Health Care and Social Assistance	3,698	4,480	4,796	1,098	2.6%
Information	1,000	1,086	1,171	171	1.6%
Management of Companies and Enterprises	194	76	42	-152	-14.2%
Manufacturing	1,589	1,419	1,352	-237	-1.6%
Mining	77	56	459	382	19.5%
Other Services, Ex. Public Admin	1,497	1,227	1,240	-257	-1.9%
Professional and Technical Services	1,548	1,563	1,641	93	0.6%
Public Administration	6,449	6,975	6,784	335	0.5%
Real Estate and Rental and Leasing	496	502	586	90	1.7%
Retail Trade	5,714	5,260	5,734	20	0.0%
Transportation and Warehousing	1,961	2,433	3,191	1,230	5.0%
Unclassified	NA	NA	NA	NA	NA
Utilities	141	150	205	64	3.8%
Wholesale Trade	766	809	1,005	239	2.8%
Total, All Industries	40,516	42,432	46,123	5,607	1.3%

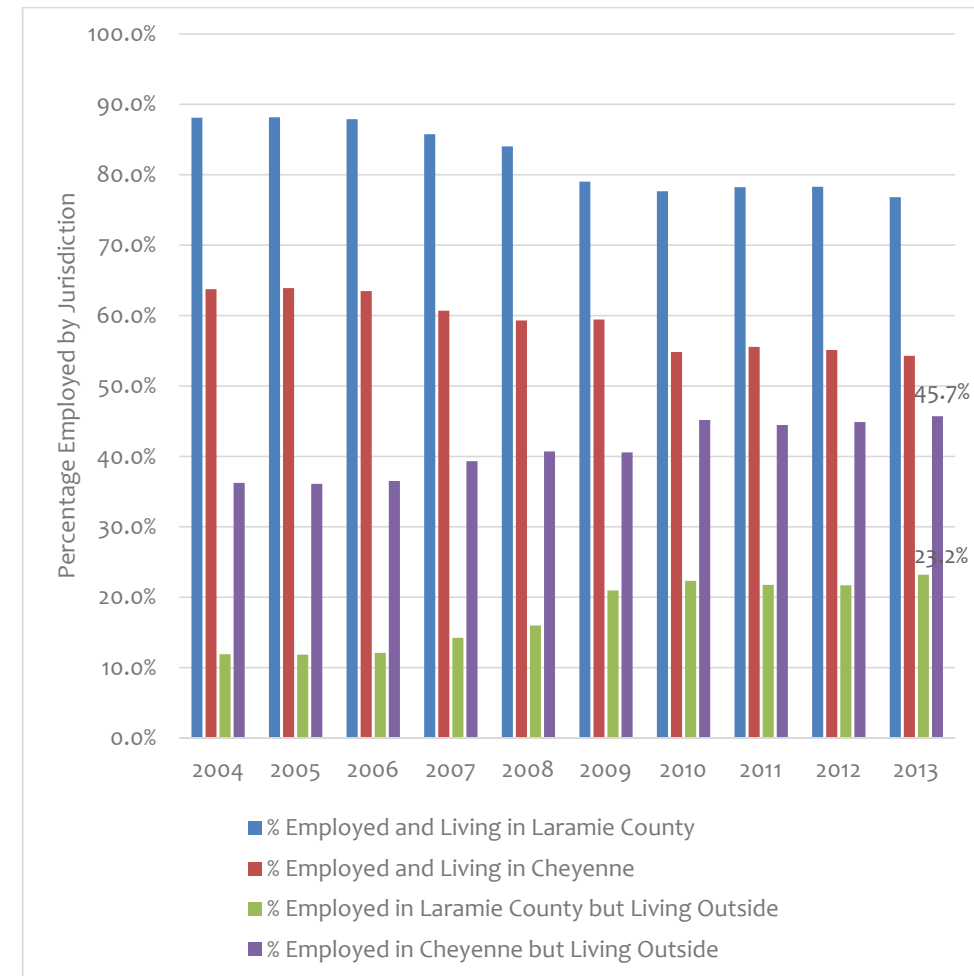
Source: US Bureau of Labor Statistics, ArLand

[1] Compound Annual Growth Rate

2.4 Commute Patterns

Most of those employed in Laramie County and Cheyenne also live within the county and city, respectively. However, the general trend over time, as seen in Figure 6, is that the percentage of those employed in Laramie County but living outside of the county has increased over time. The same is true for the percentage of jobs in Cheyenne filled by those who aren't residents of the city. The most recent figures available are from 2013, which show that about 23% of those employed in Laramie County don't live in the county. And, almost 46% of those employed in Cheyenne don't live in the city.

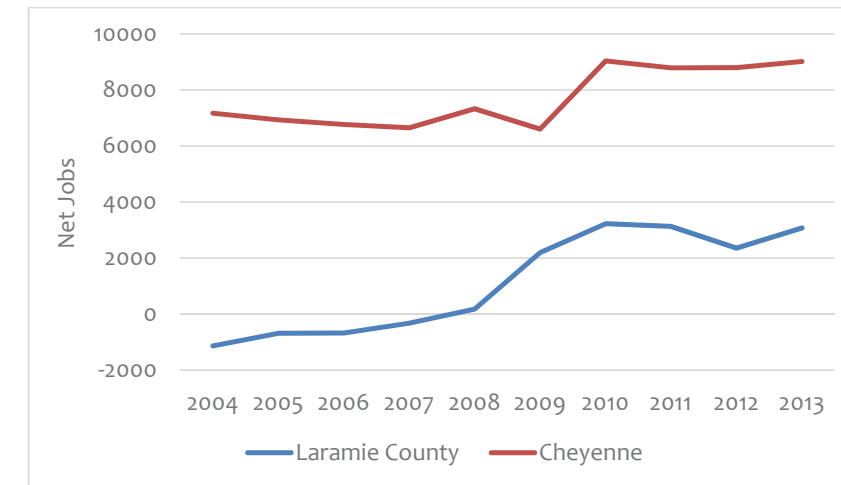
Figure 6
Laramie County and Cheyenne Employment Efficiency



Source: US Census LEHD, ArLand

The difference between the number of people employed in Laramie County or Cheyenne and the number of people living in each location is the net jobs inflow or outflow. For example, if more people are employed in Laramie County than live there, there is a net job inflow. As seen in Figure 7, Cheyenne has seen a net inflow of jobs over the past 10 years, with an increase since the Great Recession. Laramie County, on the other hand, had a net outflow of jobs until 2008. Since then the net inflow of jobs has increased from 180 in 2008 to 3,074 in 2013.

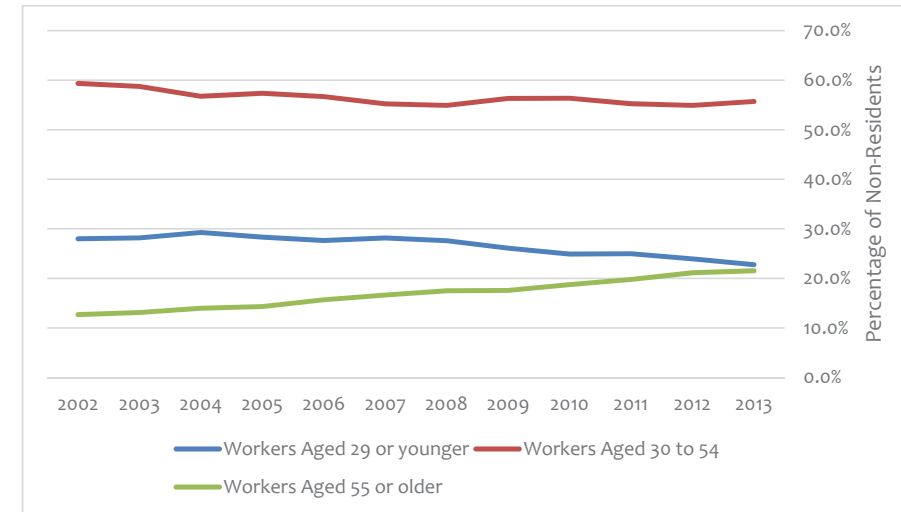
Figure 7
Net Job Inflow (+) or Outflow (-)



Source: US Census LEHD, ArLand

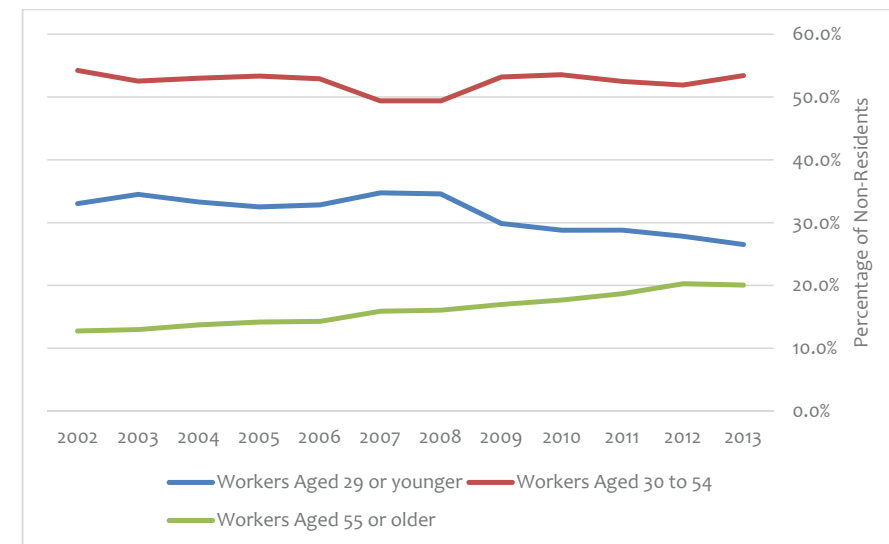
Over time, there have been changes in the age and income distribution of those people who are employed in Laramie County and Cheyenne, but live outside each respective jurisdiction. In particular, the percentage of workers age 55 or older has increased while those aged 29 or younger has decreased over time, as seen below. In addition, the percentage of non-residents working in each respective jurisdiction earning more than \$3,333 per month has increased over the past ten years, whereas the lower income categories have both seen percentage decreases.

Figure 8
Age Distribution of Non Residents Employed in Cheyenne



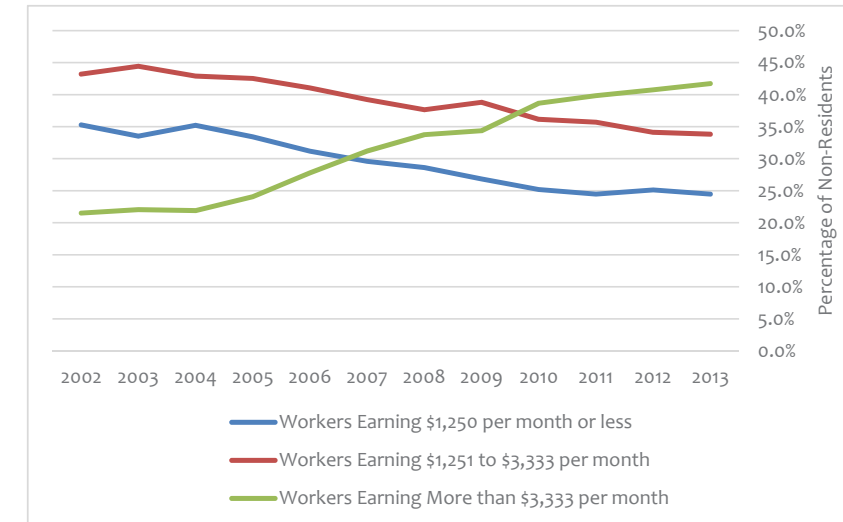
Source: US Census LEHD, ArLand

Figure 9
Age Distribution of Non Residents Employed in Laramie County



Source: US Census LEHD, ArLand

Figure 10
Income Distribution of Non Residents Employed in Cheyenne



Source: US Census LEHD, ArLand

Figure 11
Income Distribution of Non-Residents Employed in Laramie County



Source: US Census LEHD, ArLand

Where Do Non-Resident Workers Live?

Of those people who work in Laramie County, the great majority -- about 77% of them -- also live in the county. The top outside home locations in 2013 were in northern Colorado (Weld and Larimer Counties) and nearby counties in Wyoming (Albany and Natrona). Albany County is home to Laramie and Natrona County is home to Casper. This spatial distribution of those working Cheyenne is fairly similar to the county work force. It is also similar when solely considering primary jobs, rather than all jobs.

Figure 12
Where Laramie County Workers Live (2013)

	Count	Share
All Counties	46,087	100.00%
Laramie County, WY	35,399	76.80%
Weld County, CO	1,631	3.50%
Albany County, WY	1,598	3.50%
Larimer County, CO	1,252	2.70%
Natrona County, WY	1,133	2.50%
Fremont County, WY	506	1.10%
Campbell County, WY	457	1.00%
Sheridan County, WY	296	0.60%
Carbon County, WY	273	0.60%
Sweetwater County, WY	193	0.40%
All Other Locations	3,349	7.30%

Source: US Census LEHD, ArLand

Given the distribution of workers above, it’s not surprising that about 17% of those employed in Laramie County drive more than 50 miles to get to work.

Figure 13
Distance between Home and Work (2013)

	Count	Share
Total All Jobs	46,087	100.00%
Less than 10 miles	31,085	67.40%
10 to 24 miles	2,914	6.30%
25 to 50 miles	4,302	9.30%
Greater than 50 miles	7,786	16.90%

Source: US Census LEHD, ArLand

Where do Laramie County Residents Work?

Of those people who live in Laramie County but work outside the county, top destinations are adjacent counties in Wyoming (Albany and Natrona), followed by northern Colorado (Larimer and Weld Counties). Top employment centers include Laramie and Casper in Wyoming. In neighboring Colorado, Fort Collins and Greeley are top employment centers for those living in Laramie County. For those living in Cheyenne, the distribution of those working outside of the greater Cheyenne area is very similar to those living in the county.

Figure 14
Employment Location of Laramie County Residents (2013)

	Count	Share
All Counties	43,013	100.00%
Laramie County, WY	35,399	82.30%
Albany County, WY	1,187	2.80%
Natrona County, WY	1,140	2.70%
Larimer County, CO	606	1.40%
Weld County, CO	539	1.30%
Campbell County, WY	304	0.70%
Williams County, ND	251	0.60%
Platte County, WY	250	0.60%
Carbon County, WY	244	0.60%
Sweetwater County, WY	240	0.60%
All Other Locations	2,853	6.60%

Source: US Census, LEHD, ArLand

III. CASE STUDIES

The following section discusses case studies of industrial areas (characterized by railroad yards and infrastructure) that have transitioned over time to vibrant successful mixed use districts including River North (RiNo) in Denver, Colorado; the Warehouse Arts District in Lawrence, Kansas, and The Granary District in Salt Lake City, Utah.

3.1 River North (RiNo) District – Denver, Colorado

Background

The River North Art District (RiNo) is located northeast of downtown Denver and is comprised of portions of four Denver neighborhoods (Five Points, Globeville, Elyria Swansea, and Cole). The district is generally the area between Park Avenue West on the south, Interstate 70 (I-70) to the north, Interstate 25 (I-25) on the west, and Arapahoe Street to the east. The River North Area “evolved as its own community” given that it was physically cut off from downtown by the old South Platte river channel and the railroad yards. This mixed-use community included residential uses through the 1950’s, but transitioned into an industrial district over time. Industrial uses began leaving the area in the 1980’s and 1990’s, turning it into an area with high vacancy rates and deteriorating infrastructure (River North Plan 2003, RiNo Art District). Beginning in the 2000’s, the area began to turn around with a combination of private development and public planning and investment. A summary of the district’s revitalization is outlined below.

Figure 15
Land Use Map 2003



Source: RiNo Plan, 2003

One of the most important contributors to RiNo’s renewal was Downtown Denver development and revitalization, which exerted pressure on adjacent inner-ring neighborhoods. The creation of the Lower Downtown Historic District (LoDo) in 1988, helped spur public and private investment downtown throughout the 1990’s as did the construction of Coors Field baseball stadium in the mid 1990s in the area.

As neighborhoods south of RiNo, such as Ball Park and Prospect, became built out and expensive, RiNo’s proximity to downtown combined with low rents and vacant space drew artists, entrepreneurs, and creative people to the area to live and work.



Credit: Visit Denver

In 2001, a private developer with ties to the local arts community, Mickey Zeppelin with Zeppelin Development, completed a catalytic project often credited with initiating RiNo’s renaissance. Taxi I included 25,000 square feet of flexible work space for a variety of creative businesses. It was the first of many projects on the 20-acre former Yellow Cab dispatch center and corporate headquarters site on Ringsby Court. Larimer Lounge, a music venue, opened the following year, further bolstering the area’s burgeoning arts scene.

The city laid out a vision for the area in the 2003 River North Plan, which followed on the heels of transportation and access improvements to the area and the 2000 Comprehensive Plan and Blueprint Denver, which characterized RiNo as an “Area of Change.” As seen in

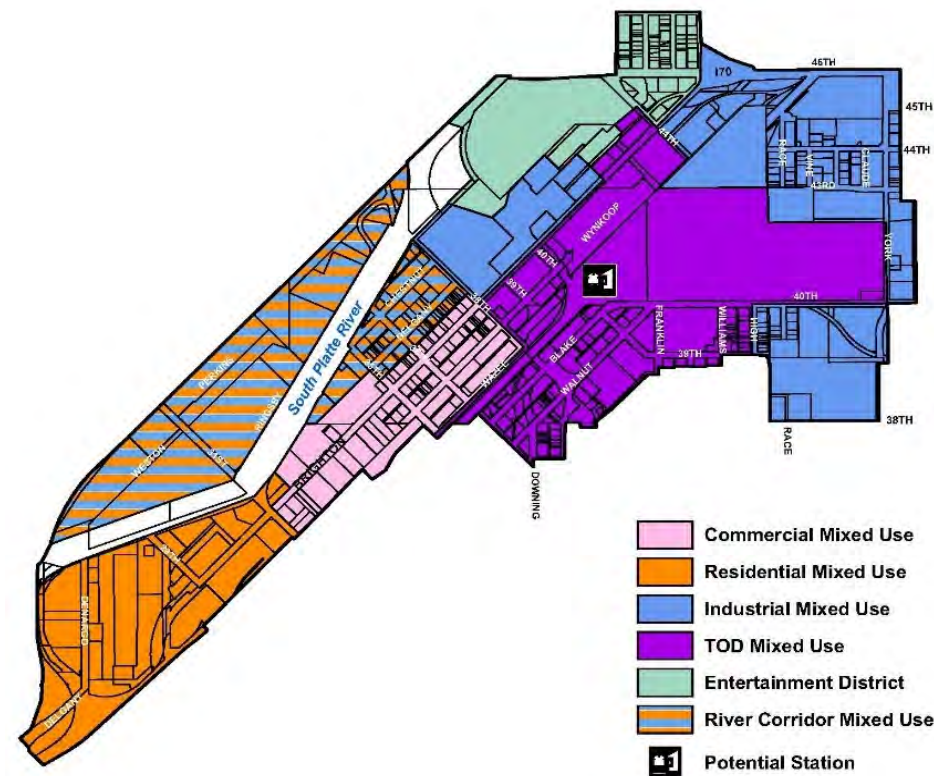
Figure 15, 60% of RiNo’s land area in 2003 was comprised of warehouses, railroad tracks and yards, and factory/food processing facilities, 9% was vacant, and a great deal of land was underutilized. Only 79 residential structures were present. In addition to promoting the area, the plan was intended to direct future growth and redevelopment in a manner that would “restore the historic, mixed-use character of the area.” As such, the plan called for instituting a new regulatory environment that makes mixed-use development possible. Figure 16 on the next page shows the new land use plan for the area.

In 2004, Denver area voters passed the FasTracks ballot initiative that solidified funding for major public transportation enhancements, giving private developers more certainty about commuter rail accessibility and a train station (38th and Blake) in the district. By 2004, several residential projects had been completed, including Brighton Flats (127 residential units), retail and commercial space, and structured parking) and Jefferson at CityGate (241 rental apartments).

The River North Arts District was created in 2005 by local artists to connect and promote the area’s artists and art scene.

The subsequent 10-year period included additional private investment coupled with public planning and investment. Two key plans for the area were adopted in 2009. The River North Greenway Master Plan sought to cleanup and develop the northern section of the South Platte River Corridor as a city amenity that would serve as a regional gateway to downtown Denver, connect neighborhoods, and create new diverse activity centers along the river corridor. That same year, the 38th and Blake Station Area Plan was adopted to guide public and private development and infrastructure

Figure 16
RiNo Conceptual Future Land Use Map



Source: RiNo Plan, 2003

investment decisions, as well as public regulation of the built environment surrounding the planned transit station.

In 2010 the city adopted a new zoning code with RiNo falling into a variety of industrial and mixed-use zone districts. The adopted zoning for the area reflected the land use pattern conceptualized in the 2003 River North Plan, (shown in Figure 16), with a higher concentration of residential and commercial development on the south end of the district and industrial and entertainment uses on the north end.

Private development and the number of new businesses and residents continued to increase as the economy recovered from the Great Recession. As an indicator of the pace of development, approximately 34% of all businesses in the district have been started since 2011 (UCD 2016). After being certified a Colorado Creative District in 2014, RiNo voters approved two special district financing tools in 2015, a General Improvement District (GID) and a Business Improvement District, to fund, among other things, additional infrastructure improvements and district-wide marketing. RiNo continues to redevelop at a frenzied pace. At the end of 2016, the 38th and Blake Street Station was an operating stop on the commuter rail line, and there were an estimated 96 developments or renovations planned in RiNo, including 5,000 residential units and 1.8 million square feet of office space.



Credit: Visit Denver

RiNo has and continues to change very rapidly. It has transitioned from an industrial district to a more mixed-use neighborhood with art galleries, restaurants, breweries, and other creative uses located in redeveloped and new buildings. The current challenge is retaining the area's grit and affordability (the key ingredients that made it attractive to artists and entrepreneurs in the first place) in the face of skyrocketing real estate values.

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3.2 Warehouse Arts District, Lawrence, Kansas

Background

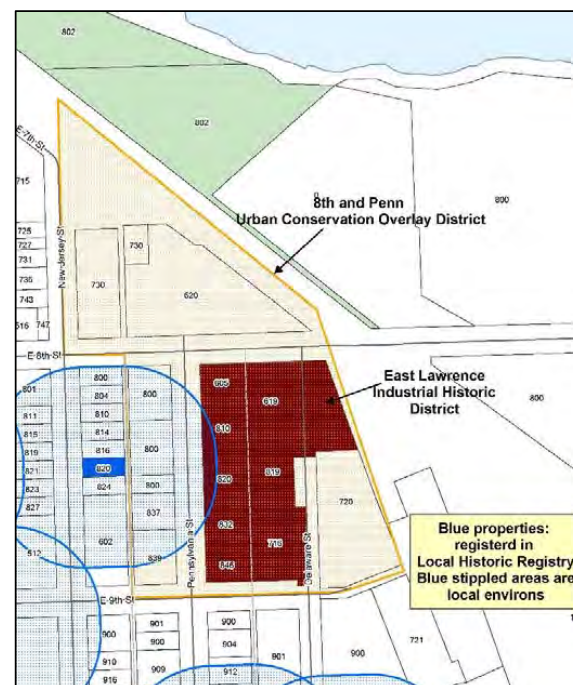
The area now referred to as the Warehouse Arts District in Lawrence, Kansas, is situated on the east side of the East Lawrence neighborhood, which is immediately east of Downtown Lawrence, which is a thriving, popular downtown. Its boundaries are those originally delineated by the City of Lawrence in 2006 as the 8th and Penn Neighborhood Redevelopment Zone, which is the same as the Urban Conservation Overlay District boundary shown in Figure 17. The southern end of the district is from East 9th Street on the south to East 8th Street on the north, and from just west of Pennsylvania Street on the west to the railroad tracks on the east. The northern portion of the district is bounded by East 8th Street on the south and New Jersey Street on the west; the railroad tracks form the northern and eastern boundary.

Neighborhood Planning

The East Lawrence Industrial Historic District (historic district) forms the heart of the Warehouse Arts District and lies within the overlay district boundary. Buildings in the historic district are “a grouping of nineteenth and early twentieth century masonry manufacturing, processing, wholesale distribution, and warehouse buildings and structures that range from one story to four stories in height and date from the early 1880s through the 1920s” (National Register Registration Form 2007).

Its transition from an underutilized commercial/industrial district with a storied past into a vibrant mixed-use district began to gather steam in the mid-2000’s. Both public planning and public and private investment were involved in implementing change, which is summarized below.

Figure 17
8th and Penn Urban Conservation Overlay District Map



Source: City of Lawrence

The desire to preserve the area likely began prior to 2000, although the East Lawrence Neighborhood Revitalization plan, approved in 2000, stated “the neighborhood believes that these [heavy industrial/commercial] areas are integral to the character of the neighborhood and should be maintained and preserved,” (East Lawrence Revitalization Plan 2000).

In April 2006, the City of Lawrence adopted its land development code. Later that year, with the goal of revitalizing the area and merging old and new land uses, the City of Lawrence developed design guidelines for the 8th and Penn Neighborhood Redevelopment Zone. The redevelopment concept outlined in the design guidelines was to mix uses both horizontally and vertically and to limit retail development to preclude “big box” retail in the area. Based on recommendations in the design guidelines, ordinances were passed at the end of 2006 that amended zoning in the 8th and Pennsylvania Street area. The 8th and Pennsylvania Street Urban Conservation Overlay District was established and General Industrial and Intensive Industrial zone districts were rezoned to a Commercial Strip zone district.

In the spring of 2007, the Neighborhood Revitalization Plan (required by the Kansas Neighborhood Revitalization Act) was adopted for the 8th and Pennsylvania District, which designated the area as a Neighborhood Revitalization area, and created a Neighborhood Revitalization Fund. In doing so, the city reduced what was deemed a “principal barrier” to revitalization and redevelopment of the district, namely costs associated with repairing, replacing, or constructing adequate public and private infrastructure. Such costs were credited as the reason qualified developers were seeking out opportunities outside the district. The ability to provide tax rebates for improvements made to eligible properties resulting in large enough increases in appraised valuation provided the financial spark that set the wheels in motion. As foreshadowed in the design guidelines, the locally significant East Lawrence Industrial Historic District was listed on the National Register of Historic Places in 2007, with five contributing buildings.

Influence of the Arts

With one of the highest percentages of working artists in the country per capita, the influence of the arts community in helping to create what the district is today should not be understated. Artists and creative business had moved into the area long before public redevelopment activities commenced. The Lawrence Arts Center, Downtown Lawrence Inc., and the Lawrence Cultural Arts Commission, along with artists, galleries, and local businesses started the Final Fridays art festival in the summer of 2010. The creation of a Cultural District in February 2013 by the City of Lawrence further solidified the area’s formal ties to art and culture.

The Cultural District is larger than the Warehouse District, generally encompassing downtown and the East Lawrence neighborhood. Its purpose is, among other things, to put Lawrence on the map as a cultural destination and to market and promote the area to enhance the city’s historic and cultural identity and promote economic activity.

Catalytic Redevelopment

In late 2011, the City of Lawrence entered into a development agreement with Tony Krsnich, a local developer, for the redevelopment of the historic Poehler Building into 49 residential units with an affordable housing requirement. The Poehler building redevelopment is credited as being the “catalyst” for the Warehouse Arts District.



Credit: Rosemann & Associates PC

Prior to 2011, zoning for this lot was changed to allow for higher residential density to accommodate the Poehler Lofts. The City agreed to pay for and implement a variety of infrastructure improvements, including storm water improvements, replacement of an inadequate water line, and streetscape and parking improvements. The project was completed in August 2012, with associated public investment of just over \$1 million (Lawrence Economic Development Office). This project utilized Low Income Housing Tax Credits for the affordable units (most of the units in the building are affordable), State/Federal Historic Tax Credits, and tax rebates. That same summer, SeedCo Studios (art and music studios) opened at 826 Pennsylvania.

This led to additional private development by Tony Krsnich that was combined with public infrastructure investment. The Cider Gallery opened in the spring of 2013, a project that utilized historic tax credits and tax rebates. Public infrastructure investment (street reconstruction, intersection improvements, parking, utility and safety improvements) of almost \$700,000 was associated with this project. The building is now a 5,000-square foot event space/fine art gallery with coworking office space. Office and studio space followed in The Poultry Building and The Annex at 720, the building SeedCo Studios (art and sound studios) moved to in 2014. In June 2015, 9 Del Lofts came online, which is a 43-unit residential project (36 affordable) that was jointly developed with Tenants to Homeowners (a community land trust). This further expanded the number of residents in the Warehouse Arts District. This project also utilized tax reduction tools and public investment, including Low Income Housing Tax Credits, HOME Funds, AHP Funds, tax rebates, and about \$270,000 in public infrastructure investment (relocation/connection of sanitary sewer main and water line, public trail, and burying utility lines) (Lawrence Economic Development Office).

Current Activities

The City of Lawrence is currently engaged in a planning process for the 9th Street Corridor, which will fuse public art into a streetscape project that will also include multi-modal transportation strategies along East Ninth Street, from Delaware to Massachusetts. Once implemented, this project will enhance the connection between downtown, and the Warehouse Arts District. Although much of the private investment in the area has been completed by Tony Krsnich (now Flint Hills Holdings), he has sold 826 Pennsylvania to another developer currently working on a brew pub and apartments in the Kansas Seed Co. building.

The story of the Warehouse Arts District’s revitalization is one of a marriage of public and private investment. In this case, public design guidelines, zoning changes, and tax rebates were integral to the area’s redevelopment. Limitations on retail development has helped keep the area from becoming solely a retail district, and the National Register listing has assisted in retaining the area’s

historic fabric. The ongoing challenges for the district will be preserving the area’s industrial and rail roots, retaining close ties to local artists, and maintaining residential and commercial affordability.

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3.3 The Granary District, Salt Lake City, Utah

Background

The Granary District is located immediately south of downtown Salt Lake City, Utah. As seen in Figure 18, it is generally the area from 900 South to 600 South, and from Interstate 15 (I-15) to 300 West, encompassing about 96 acres.

Historically, the larger Gateway District of Salt Lake City, of which The Granary district is a part, saw industrial and transportation uses begin to take the place of residences when the railroad arrived. Neighborhoods comprised of various ethnic groups formed with the arrival of the railroads, but disappeared after World War 1. The construction of I-15 created an area “occupied by railroad tracks, highway ramps, overpasses, and heavy industrial uses,” which began to deteriorate and become a haven for crime (The Gateway Specific Plan 1998). In the late 1990’s, the Gateway District was characterized as an underutilized area with “[a] mix of low-intensity uses, warehousing, general commercial, distribution, and light industrial uses” (The Gateway Specific Plan 1998). The revitalization of the southern portion of the Gateway District, now called The Granary, is the result of public planning processes and public and private investment. The discussion below summarizes the area’s transition from an underutilized industrial district to a more vibrant mixed-use district that is home to restaurants, coffee shops, residential units, and active, creative businesses.

Neighborhood Planning

The history of the area’s change can at least be traced to 1978 when a local community group proposed studying the revitalization of the “Gateway.” A Visionary Gateway Plan was completed in 1994, a Downtown Plan in 1995, and a Rail Consolidation Plan in 1997. The consolidation of the railroad tracks in the area, the shortening of bridges over the railroad tracks, the relocation of Amtrak, and the 2002 Winter Olympic Games were factors driving change, among others. The Gateway District Land Use & Development Master Plan and Gateway Specific Plan, both adopted in 1998, still guide the area’s redevelopment. These plans called for a mix of uses in the district, with housing being a key recommendation, including above commercial and other uses when appropriate.

Figure 18
Granary District Project Area



Source: Redevelopment Agency of Salt Lake City

Granary District Redevelopment Project Area Designation

In addition to the vision outlined in the plans mentioned above, the Redevelopment Agency of Salt Lake City (RDA), originally formed in 1969, has played a central role in The Granary redevelopment. In April 1999, the RDA created the Granary District Redevelopment Project Area after finding the area to be blighted in accordance with the Utah Neighborhood Development Act, as amended. The intent of the agency has been to remove blight from the area through public infrastructure, including flooding mitigation, improvements, building rehabilitation and renovation, and new development, which includes the ability to assist with financing through loans, grants, and tax incentives (tax increment).

Design Vision: “The scale of development ranges from large scale multifamily and commercial uses to single family housing on less than 5000 SF lots. The vision for new development in this area is to create a walkable, transit-oriented neighborhood by fostering the continued pattern of small scale infill projects as well as allowing for context sensitive larger scale mixed-use projects that will provide a wide range of housing choices and incubator commercial spaces that serve the neighborhood and the region” (SLCRDA)



Credit: artspaceutah.org

Catalytic Project

The RDA was first involved in an office renovation in the district in 2000. It has provided financial assistance in the district for specific buildings and street improvements, including two renovation loans and tax increment reimbursement for the Artspace Commons project at 824 400 West. This project developed by Artspace, a non-profit developer, came online in the fall of 2010 and is generally viewed as a catalytic project for the area. It involved cleaning up a designated brownfield site (a former scrap metal recycling and storage lot) and building a mixed-use LEED (Leadership in Energy & Environmental Design) Certified (Gold) housing project. The project consists of 102 one and two-bedroom affordable apartment units and 50,000 square feet of commercial space (artist-studio, office, and retail space). The project utilized \$27.1 million in New Market Tax Credit financing and tax increment dollars through the RDA, as well as a \$50,000 grant from Enterprise’s Green Communities initiative, and small permanent loans from the Utah Community Reinvestment Corporation (Artspace Utah and Enterprise).

In 2013, Artspace also developed Artspace Solar Gardens south of Artspace Commons. Artspace Solar Gardens is a net-zero, mixed-use building comprised of 30 affordable residential units and 10,000 square feet of commercial space for artist studios, office, and retail space, as well as a community garden. This project also utilized New Market Tax Credits as part of project financing.

Area TOD

Immediately east of The Granary is the Central Ninth Neighborhood, which is within the RDA’s West Temple Gateway Project Area. The RDA’s work in this project area is focused on transit-oriented development and has included funding the 900 South TRAX Station, an infill light rail station two blocks east of Artspace Commons that opened in September 2005. This coupled with beautification and infrastructure enhancements helped revitalize this neighborhood immediately east of The Granary. In addition, as early as 2008, a streetcar was and continues to be envisioned along 400 West.

RDA Redevelopment Strategy

Following the completion of Artspace Commons, the RDA approved a Redevelopment Strategy for both the Granary District and West Temple Gateway in June 2011. The strategy included a redevelopment guide, design guidelines, and site-specific design and redevelopment options for four RDA-owned properties and the Fleet Block, a large city-owned property in The Granary district. A few months later, the 9th and 9th trail/linear park opened. Although not all segments are completed, the project seeks to enhance the linkage between the east and west side of town, which further increases transportation mobility in the district and is one more area amenity.

Private Investment

Even though the area was already home to many businesses and residents, and a variety of public plans and projects had been completed, James Alfandre, a local new urbanist developer, recognized a lack of community cohesion in the area. To assist in redevelopment, he brought neighbors and business owners together online and in-person around 2011 to crowdsource the area's redevelopment, which led to the birth of The Kentland Initiative, a 501(c)3 organization. The initiative has helped brand the area, connect stakeholders, organize block parties, and has even held a Crowdsourced Redevelopment charrette that led to the idea for Granary Row, a temporary (a few months) shipping container pop-up market and festival space in the middle of the street (700 South between 300 and 400 West).



Credit: inhabit.com

As the district continues to revitalize, the RDA continues to focus on key projects and strategies, which are outlined in their 2015-2019 Strategic Plan for the Granary District. This plan prioritized street reconstruction and redevelopment, redevelopment of the Fleet Block, creation of open space, and realization of the downtown street car, which is intended to serve as a catalyst for redevelopment. The RDA recently provided forgivable adaptive reuse loans to two businesses in the district, Atmosphere Studios and Fisher Brewing Company.

The Granary District's success is not the result of one public or private plan or action, but rather a blend of both. Given that the district still has vacant and underutilized properties, public infrastructure investment has been highlighted as a needed "catalytic project" in the city's 2016 Downtown Plan to entice private investment, and to further facilitate the district's transition to an even more vibrant, walkable, and connected district that is rich in arts and culture. The challenges going forward will likely be the same as those faced elsewhere, namely affordability and retention of the grit and character of its industrial and railroad roots.

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IV. MARKET SUMMARY

The following section briefly discusses each of the potential land uses in the West Edge district. One of the challenges is that there is currently relatively little development activity in the West Edge to suggest that without public incentives and/or activity, that development / market activity is primed to come to the area.

As the case studies indicate, redeveloping warehouse districts start from a base of strong downtowns, public investment and political will. Metro areas like Denver have seen renewed interest in the downtown core and immediately surrounding neighborhoods. While downtown Cheyenne has seen some activity, strong downtown related growth and development remains elusive which is a challenge for the West Edge.

As the RiNo case study indicates, downtown Denver has also expanded over time from its core downtown to include areas in the Lower Downtown (LoDo) area. Given the growth forecast and high density developments in the planning pipeline in the RiNo area, and location relative to downtown, it is poised to become another downtown subdistrict.

The following analysis suggests that the West Edge be considered a subdistrict of downtown Cheyenne. Albeit different in form and land uses, the West Edge is poised to benefit from the same conditions benefiting downtowns throughout the country including a desire for a walkable urban environment, locally owned services and amenities and downtown residential living options.

4.1 Residential

While there have been multifamily units built over the years, the majority of residential units built in Laramie County have been single family detached units as seen in Table 6. Recent planning efforts examining housing demand in Cheyenne, however, have pointed to a strong desire for greater diversification of housing stock available in the area.

Table 6
Residential Construction Activity

Residential Construction Activity	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
Laramie County											
Single Family	674	460	321	203	229	550	277	350	424	313	370
Duplex	0	2	0	0	2	0	0	0	0	0	0
Tri and Four Plex	44	0	0	44	0	0	0	0	16	4	68
Multifamily	132	8	0	6	216	0	0	132	156	0	0
Total	850	470	321	253	447	550	277	482	596	317	438

Source: WCBEA from City of Cheyenne, Laramie County, ArLand

The desire for housing diversification is partially driven by housing affordability. Figure 19 and Figure 20 show the ever increasing rental and sales prices for single family detached housing in the Cheyenne market area. In general, the rates of increases for single family detached homes have

outpaced increases for apartments and mobile home units, making attached housing options increasingly attractive.

Figure 19
Cheyenne Housing Rental Rates

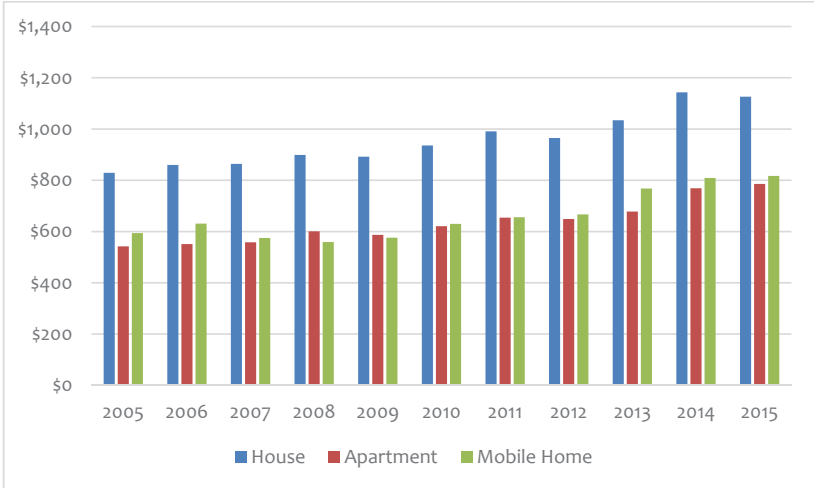
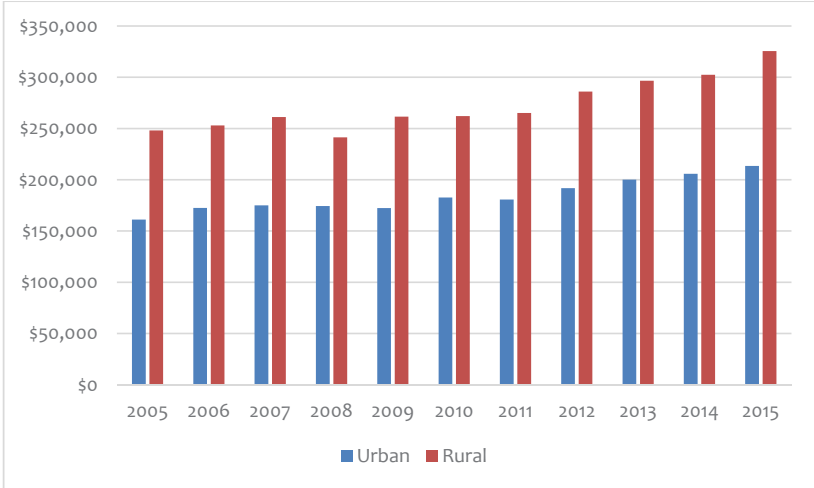


Figure 20
Cheyenne Average Housing Sales Prices



Source: WCBEA, ArLand

The Cheyenne and Laramie County Housing Assessment completed in 2015 by the Rural Resource Team indicated the lack of quality housing options and affordable housing options in Cheyenne. It also indicted a desire for downtown lofts, senior housing and other residential types appropriate for more urban settings.

Table 7
Residential Demand for the West Edge / Reed Avenue

Cheyenne Households 2017	27,853
Cheyenne Households 2027	30,797
HH Growth	2,944
Vacancy Rate	5.00%
Demolition Rate/yr.	5.00%
Total Unit Requirement	3,246
Current Downtown Housing Units	630
% of Overall Market	2.3%
2.3% of Future Households	73
5% of Future Households	162

Source: WCBEA, ArLand, CoStar

Table 7 shows an estimate of 10 year demand for West Edge / Downtown residential housing units. There are currently an estimated 630 housing units in the broader West Edge / Downtown area with most of them in a multifamily (attached, urban) type of configuration. One of the market challenges is that there hasn’t been a significant increase in urban living in the downtown, with the exception of a few units on second floors, in the downtown area for a number of years. However, assuming that the West Edge / downtown are able to capture their current share of future growth would result in estimated demand for an additional 73 units in 10 years for the West Edge / downtown area.

Given the case studies as well as community demand, it would be anticipated that a significant portion of residential demand that could be met in the West Edge would be affordable. As shown in the case studies, affordable housing can have a catalytic effect on area redevelopment. However, further analysis would be needed to determine the level of affordability, potential financing, etc. Given the lack of affordable, quality, multifamily housing options in the area, tax credit housing could be pursued for project financing and to help fill an affordable housing need.

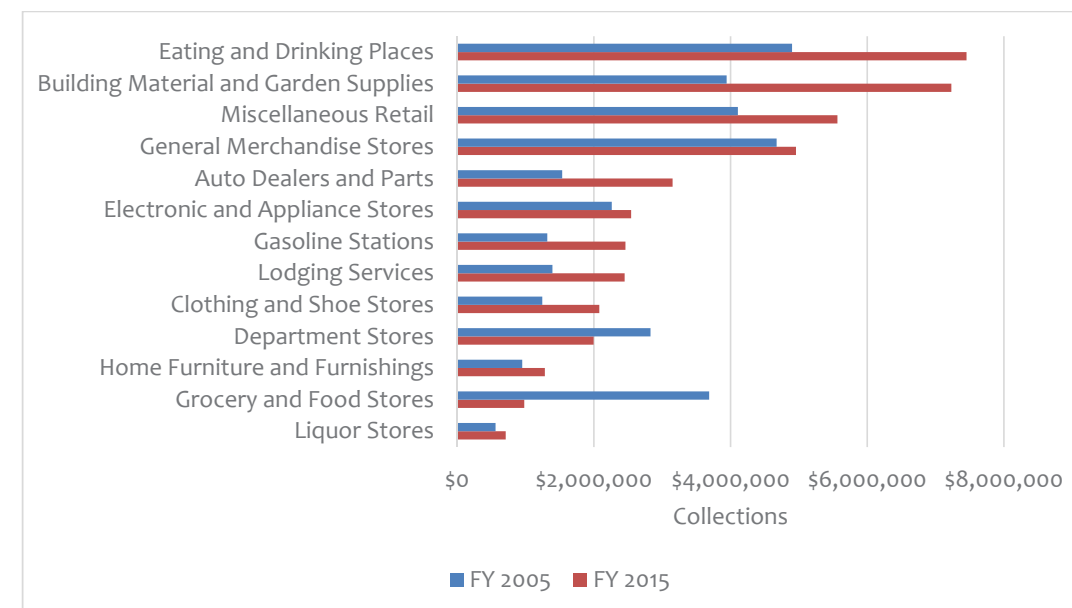
It should be noted that Arts Cheyenne has been working with an arts-based developer who has identified the potential for 30 affordable housing units for artists in the West Edge / Downtown area, along with creative space. There is at least one property owner experimenting with adding a few residential units to their mixed use development. If initial residential development is successful, up to 162 residential units could be supportable in 10 years. This would require some density in key locations.

4.2 Retail

Interviews for this plan have indicated a desire for unique, boutique retail, restaurants, including microbreweries, and a number of interesting, locally owned retail options. Statistics examined in this report and elsewhere have cited the commuting patterns of area employees. Interviews indicate that while jobs may be in Cheyenne and Laramie County, that quality of life indicators encourage employees to live outside of the area in areas with “quality of life”. While affordability may be an issue, arguably more important considerations are these “quality of life” factors, which include activities, interesting, unique places to visit, eat and recreate.

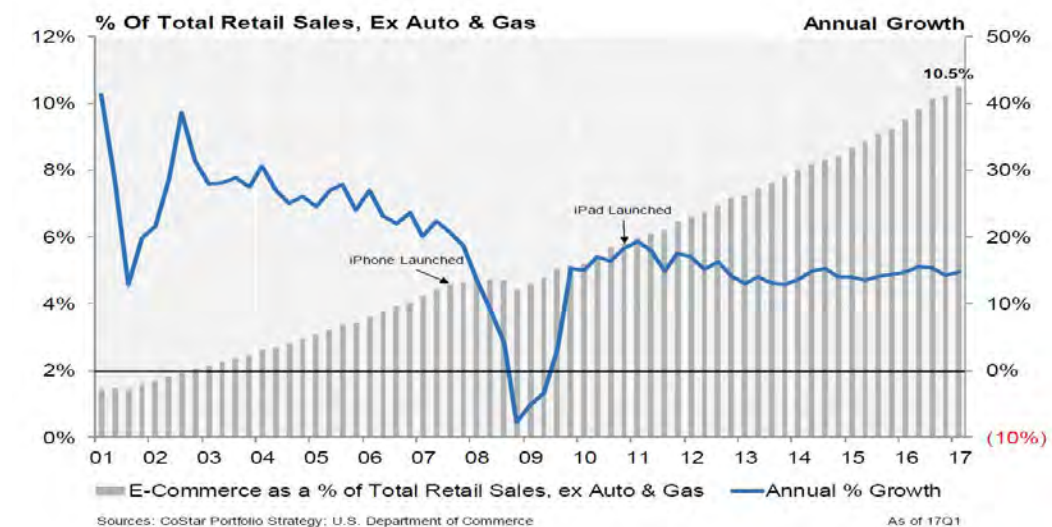
One interesting phenomenon taking place in Laramie County and nationwide is the increased revenues for Eating and Drinking Places (restaurants), especially when compared to grocery and food store revenues. Sales tax collections are a good proxy for analyzing this trend (Figure 21). While there are certainly reasons for decreased collections at grocery stores (increased grocery spending at General Merchandise stores including Wal-Mart), it points to a change in how people spend money and time. It also points to potential demand for eating and drinking establishments in the Cheyenne area.

Figure 21
Laramie County Sales Tax Collections FY 2005 and FY 2015



Source: Laramie County, ArLand

Figure 22
E-Commerce as a Percentage of Total Retail Sales



Source: CoStar

Retail is challenging in today’s environment, because of nationwide trends including the increased ease and popularity of e-commerce. It is estimated that 10.5% of all retail sales nationally are via the internet and this percentage continues to increase (Figure 22). Restaurant spending is displacing retail spending as patrons desire “experiences” rather than “things”.

Because Cheyenne and Laramie County are continuing to grow, there will continue to be demand for some commercial services because of the growth in population and households. However, initial retail offerings in the West Edge should be those that offer experiences, such as microbreweries, interesting / unique restaurants, entertainment, etc. In the long term, if the West Edge starts to gain traction, unique clothing, furniture and other specialty items would be supportable, however, they are uses that are more appropriate in the long term, after the district is more established.

Table 8 shows the potential demand for the retail store types most appropriate for the West Edge / downtown area. Based on the growth in population and households, average household incomes, and how people spend their money, it is estimated that there is approximately 15,000 to 45,000 square feet of retail and restaurant demand.

Table 8
Retail Demand for West Edge / Reed Avenue

Laramie County Households 2017	98,354		
Laramie County Households 2027	108,750		
HH Growth	10,396		
Average Household Income	\$75,000		
Total Income	\$779,721,975		
Demand for:	% of Income	\$ Demand	
Grocery Stores	6.7%	\$52,158,338	
Beer, Wine, and Liquor Stores	0.4%	\$3,160,954	
Clothing, Furniture, Specialty Items	5.8%	\$45,166,935	
Restaurants	6.2%	\$48,465,488	
	\$ / SF Revenues	SF Demand	% of Demand for West Edge
Grocery Stores	\$400	130,396	13,040
Beer, Wine, and Liquor Stores	\$350	9,031	903
Clothing, Furniture, Specialty Items	\$300	150,556	15,056
Restaurants	\$300	161,552	16,155
Demand for West Edge / Reed Avenue (sf)		15,000 to	45,154

Source: ArLand, CoStar

One of the biggest challenges for the West Edge area is the fact that it is relatively untested from a market and development perspective. Many warehouse districts have evolved into restaurant and entertainment districts. One of the more popular immediate opportunity for the West Edge may be a microbrewery.

Expanding restaurant and entertainment in the West Edge may be a challenge because of the current liquor laws. While oversimplified, the following describes the alcohol situation.

- *Microbrewery:* License allows a business to serve their own manufactured beer along with other malt beverages only. The permit is \$500 per year.
- *Retail Liquor Licenses:* License allow for bars and liquor stores. The number of licenses is based on population and state statutes, and distributed by City Council. No new licenses are available until 2020. These licenses reportedly sell for \$200,000 on the black market.
- *Restaurant Liquor License:* License would enable a business to serve alcohol and food. Revenues, however, need to be at least 60% from food and no more than 40% from alcohol.

There have been microbreweries in Cheyenne which have opened with separate food trucks in order to be able to offer food. The food stipulation and the limited (and expensive) liquor licenses serve as a barrier for significant redevelopment of this use type.

4.3 Employment

The West Edge is currently primarily an employment center. Between the West Edge and downtown, there are about 840,000 square feet of office and nearly 115,000 square feet of industrial space as shown in Table 9 and Table 10. One of the challenges is that there hasn't been much recent large scale additions to the office and inventory in the downtown or the West Edge in the last 10 years. The recent employment growth in the Cheyenne market has been outside of the downtown area.

Table 9
Office Indicators

	West Edge / Downtown	Cheyenne	Laramie County
Office Total Square Feet	838,247	2,408,718	2,419,852
Current Rent per Square Foot	\$11.53	\$14.81	\$14.79
Current Vacancy Rate	5.6%	5.0%	5.0%
12 Month Absorption (sf)	7,282	23,341	24,841
Months on Market	11.9	7.7	8.0
Additions to Inventory (2007-2017)	0	181,128	189,628

Source: CoStar, ArLand

Table 10
Industrial Indicators

	West Edge / Downtown	Cheyenne	Laramie County
Industrial Total Square Feet	114,780	4,912,049	5,075,055
Current Rent per Square Foot	\$5.86	\$7.43	\$7.40
Current Vacancy Rate	10.3%	2.8%	2.8%
12 Month Absorption (sf)	13,289	4,040	-960
Months on Market	6.4	11.5	11.5
Additions to Inventory (2007-2017)	0	194,799	194,799

Source: CoStar, ArLand

Table 11
Office / Industrial Demand for West Edge / Reed Avenue

	Current Employment Uses		
	West Edge	Cheyenne	Laramie County
Office	838,247	2,408,718	2,419,852
Industrial	114,780	4,912,049	5,075,055
Total Employment Uses	953,027	7,320,767	7,494,907
W Edge % of		13.0%	12.7%
Laramie County Employment grows by 1.1% annually			
2027 Estimate of Laramie County Employment Square Feet			8,359,886
West Edge / Downtown at Current Share			1,063,015
West Edge / Downtown Employment Space (SF)		50,000 to	109,988

Source: WCBEA, CoStar, ArLand

Employment in Cheyenne and Laramie County is continued to forecast to grow by 1.1% annually. Assuming that office and industrial real estate grows at a commensurate rate and that the West Edge and downtown continue to occupy its current share of future growth, approximately 100,000 square feet of office / industrial square footage could be supportable by 2027 as shown in Table 11.

APPENDIX B

ECONOMIC & FISCAL IMPACT ASSESSMENT



September 11, 2017

To: Cheyenne Reed Avenue Team

From: Arleen Taniwaki, ArLand Land Use Economics

Re: Reed Avenue Financial / Fiscal Analysis

The potential redevelopment of the Reed Avenue Corridor presents the City of Cheyenne with an opportunity to catalyze the redevelopment of the West Edge neighborhood. Several recent planning efforts have helped to solidify the City's vision for the area. The reinvigoration of the Reed Avenue corridor is one of the first steps in implementing the vision.

This memo outlines:

- The potential fiscal impact of redevelopment of the Reed Avenue Corridor over a 25 year time horizon. The analysis includes an estimate of property and sales tax generated, as well as jobs created.
- The potential tax increment over a 25 year time horizon including all tax increment, as well as tax increment assuming Cheyenne's portion of property and sales taxes alone. It also assumes that 20 mills for commercial projects are levied on properties currently in the Cheyenne Downtown Development Authority boundaries.
- Strategic steps related to special districts and grant programs which are part of a larger strategy for the City, Metropolitan Planning Organization and its partners.

The financial / fiscal analysis assumes that area improvements result in economic development projects.

- Figure 1 depicts the Reed Avenue corridor and property redevelopment assumptions. In addition to maintaining some of the current industrial land uses, other future land uses include Office Flex, Commercial / Retail / Restaurants, Townhome, and Mixed Uses. The land use mix is as follows:
 - 180,000 square feet of office / flex / employment
 - 68 townhomes
 - 35,000 square feet of commercial / retail / restaurant uses
 - 110,000 square feet of "mixed uses" which could include a combination of employment and residential uses
- Property development values were assumed as follows:
 - Townhomes valued at \$190,000 per unit
 - Office / Flex valued at \$100 per square foot
 - Commercial and Mixed valued at \$150 per square foot
- Figure 2 shows the properties that were assumed for the Reed Avenue "District".

Figure 1
Reed Avenue Redevelopment Parcels

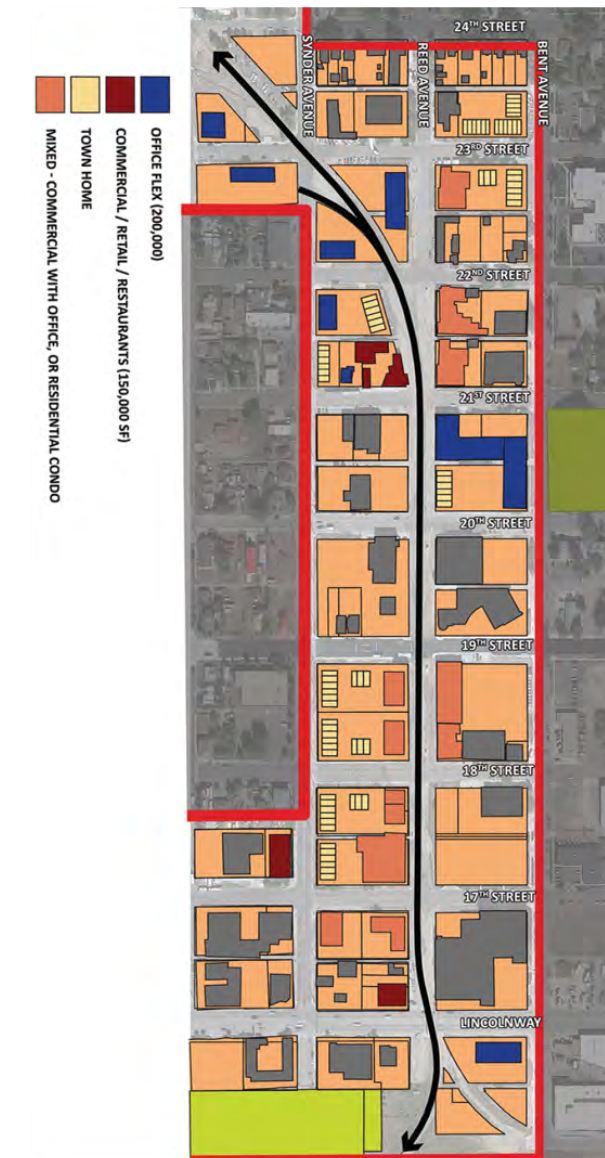


Figure 2
Reed Avenue “District”



FISCAL IMPACT

The overall fiscal impact of redevelopment examines the overall benefit related to area redevelopment. It assumes that everyone in the community benefits from the property taxes (at 72.2 mills+20 mills for commercial DDA properties) and the sales taxes (4% state / local and 2% local) that are levied. The benefits include an increase in property tax collections as well as property and sales taxes for new commercial projects in the area. They are outlined below.

Current	
Current Annual Property Tax Generated	\$203,000
New Property Tax Generation [1][2]	
Annual Property Tax	\$203,000 increasing to \$943,000 annually
Cumulative Property Tax (25 years)	\$15.2 Million
New Sales Tax Generation [2,3]	
New Sales Generation	\$16 Million to \$20 Million annually
New Sales Tax Generation (state and local)	\$1 to \$1.2 million annually
Local Sales Tax Generation	\$370,000 to \$450,000 annually with cumulative total of \$4 million
New Jobs	540

- [1] Assumes that property redevelopment commences in 2022
[2] Assumes a 2% average annual increase in property and sales values from 2017.
[3] Assumes sales are generated at approximately \$350 per square foot with 35,000 square feet of commercial space. Development starts in 2032.
[4] Sales tax generation (4% state / local and 2% local only)

Because the Reed Avenue Corridor currently generates some property taxes, the table below summaries the “net new” property and sales tax generated. It assumes that 35,000 square feet of commercial (sales estimated at \$350 per square foot) generates new sales tax revenues for the area.

TAX INCREMENT (TOTAL)

Current	
Current Annual Property Tax Generated	\$203,000
New Property Tax Generation [1][2]	
Annual Property Tax Increment	\$4,000 to \$740,000 annually
Net New Cumulative Property Tax	\$9.3 Million
New Sales Tax Generation [2,3]	
New Sales Generation	\$16 Million to \$20 Million annually
New Sales Tax Generation (state and local)	\$1 to \$1.2 million annually with cumulative total of \$11 million

The total below shows just the calculation based on Cheyenne’s portion of property taxes (8 mills) and its portion of sales taxes (Cheyenne receives 68.37% of 31% of the 4% statewide sales tax. Cheyenne also receives 68.37% of a 2% local sales tax).

TAX INCREMENT (CHEYENNE ONLY)

Current	
Current Annual Property Tax Generated	\$46,000
New Property Tax Generation [1][2]	
Annual Property Tax Increment	\$930 to \$85,000
Net New Cumulative Property Tax	\$1.0 Million
New Sales Tax Generation [2,3]	
New Sales Generation	\$16 Million to \$20 Million annually
Local Sales Tax Generation	\$370,000 to \$450,000 annually with cumulative total of \$4 million

STRATEGY

The following steps are part of larger strategy which would include working with the railroads, prioritizing and funding local projects, building political support, working with the local business community, etc. Steps would also include:

- Work with DDA on expanding its boundary to include the entire Reed Avenue corridor
- Through City / MPO / DDA efforts, pursue additional grant funding as identified by Ayres and Associates, October 1, 2016. Sources include (as identified)
 - Greening America’s Communities: US EPA
 - TIGER: US DOT
 - Local Foods, Local Places: US EPA
 - EDA Planning Program and Local Technical Assistance: US Department of Commerce
 - Cool / Connected – EPA and USDA Program
 - EDA Public Works and Economic Adjustment Assistance: US Department of Commerce
 - Community Development Block Grant
 - Cheyenne-Laramie County Economic Development Joint Powers Board
 - Business Committed Grant: Wyoming Business Council
 - Community Readiness Projects: Wyoming Business Council
 - Community Enhancement Grant: Wyoming Business Council
 - WYDOT Transportation Alternatives: WYDOT
 - Business Ready Community Planning Projects – Wyoming Business Council

DRAFT

REED AVENUE
RAIL CORRIDOR
MASTER PLAN

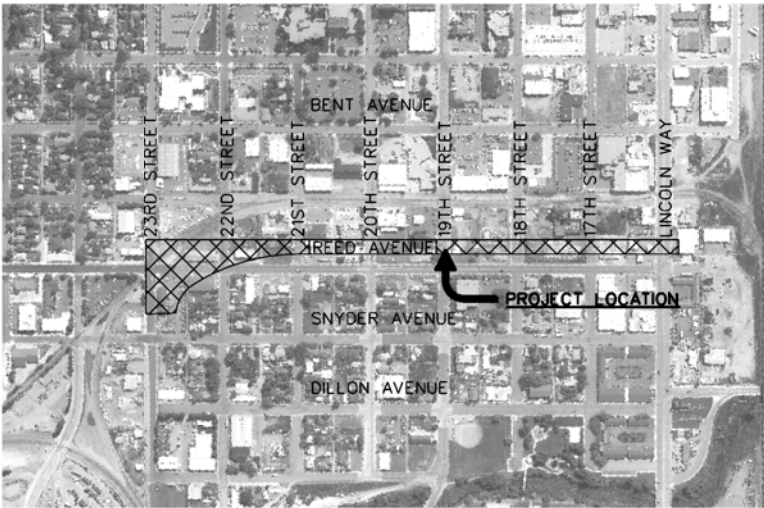
APPENDIX C

CONCEPTUAL CORRIDOR ENGINEERING

CITY OF
CHEYENNE, WYOMING
CONCEPT PLANS
FOR
REED AVENUE
RAIL CORRIDOR

PROJECT LOCATION

- CONCRETE PAVEMENT (PLAZA)
- CONCRETE SIDEWALK
- CONCRETE CURB & GUTTER
- LANDSCAPE AREA
- PAVERS
- EXISTING RAILROAD CROSSING
- ON-STREET PARKING LANE
- BICYCLE LANE
- FENCE
- TREE
- R.O.W.
- RAILROAD SIGNAL
- RAILROAD SIGNAL CABIN



NOTE:
THE EXACT LOCATION OF UNDERGROUND UTILITIES SUCH AS GAS, TELEPHONE, FIBER OPTIC, ELECTRIC, CABLE TV, AND PIPE LINES ARE UNKNOWN. THE CONTRACTOR SHALL CONTACT DIGGERS HOTLINE AT 811 AND ALL OTHER UTILITY OWNERS WHICH ARE IN THE PROJECT LIMITS, BEFORE COMMENCING EXCAVATION.

THE SUBSURFACE UTILITY QUALITY INFORMATION IN THIS PLAN IS LEVEL D. THIS UTILITY QUALITY LEVEL WAS DETERMINED ACCORDING TO THE GUIDELINES OF C/ASCE 38-02 ENTITLED "STANDARD GUIDELINES FOR THE COLLECTION AND DEPICTION OF EXISTING SUBSURFACE UTILITY DATA."



INDEX

SHEET NO.	DESCRIPTION
1	TITLE SHEET
2	TYPICAL SECTION
3-9	PLAN SHEETS

DECEMBER 06, 2017

CHEYENNE, WYOMING

FILE NO.

01
09

